

Score-linked practice intelligence from long-form violin training video

Alan N. Pham¹

¹AO Labs, Worcester, MA, USA

High-level music training produces dense evidence, but daily practice videos rarely become a usable longitudinal record. We report Curtis, a live AO Labs system that converts long-form violin practice media into source-linked practice records: active-playing windows, detected audio fragments, repertoire labels, score-alignment candidates, source-score review queues, gold review labels, and accepted/rejected evidence gates. At the 15 May 2026 evidence lock, the system indexed the public @nalalan YouTube channel, separated uploaded-video duration from detected active playing, and required explicit agreement between audio, transcription, parsed source score, visible source crop, and musician review before showing any score-linked phrase as accepted evidence. The current strict state intentionally returns the rejected Scherzo-Tarantelle phrase lane to zero accepted score-linked phrases. The contribution is not an admissions prediction or a solved transcription benchmark. It is a falsifiable method for turning messy practice video into inspectable musical evidence while preserving uncertainty, rejecting wrong matches, and making future practice, score matching, and feedback claims auditable.

1 Introduction

Elite music preparation depends on repeated, deliberate practice and feedback over long time horizons (1–3). The useful data are not only polished performances. They include ordinary practice sessions, failed takes, camera angle limitations, recurring technical problems, and changes in focus across days. A violinist recording daily practice can therefore produce a large longitudinal dataset, but the raw dataset does not automatically become an operating record.

Curtis addresses this problem as a live software system rather than as a portfolio page (4). The system indexes public practice media, separates uploaded video duration from detected violin-playing practice time, records technical observations, and keeps evidence tied to clips and generated notation. It is designed for the Curtis Institute goal, but it does not claim to estimate an acceptance probability. The current role of the system is narrower: observe practice evidence, retain history, separate judged from unjudged evidence, and identify the specific passage-level blockers that can be tested in subsequent recordings.

The core pipeline is video → active practice detection → clips → music transcription → repeat grouping → score matching → heat map → repertoire update → passage-level observations. It is built against established difficulties in onset detection, pitch estimation, automatic music transcription, and music information retrieval rather than assuming that a video can be cleanly converted into a score (5–9). The output is intentionally small: analyzed practice days and an automatically updated repertoire list, both grounded in clips, notation, active playing time, and confidence limits.

The system is also part of AO Labs’ broader progress architecture. In May 2026, AO Progress was introduced as a shared ledger for cross-application state and long-term changes across public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state (10). Curtis supplies one stream in that ledger: timestamped media and review evidence for musical development.

2 System state

The production state uses the live backend at <https://curtis.aolabs.io/api/curtis/media-status>, public YouTube metadata, MusicXML-derived score parsing, and local source-score assets where available (11–13). The checked source state at 2026-05-15 05:30 PM EDT reported the values in Table 1. These are operating facts, not outcome claims.

A later implementation check at 2026-05-14 08:29 AM EDT corrected and extended the Scherzo-Tarantelle score-truth source, but subsequent review showed that the intermediate D C Bb opening map was still a staff-position error rather than accepted source truth. The previous D A# G D source-score phrase, the later D C Bb D C Bb D source map, and the five-note A#4 D5 C5 A#4 D5 / Bb4 D5 C5 Bb4 D5 display lane are all rejected as accepted symbolic-score evidence. The 2026-05-14 07:04 PM EDT actual-source score-gate update added an exact-range source crop for that five-note subrange and prohibited generated notation SVGs from appearing as accepted score snippets. The 2026-05-14 09:45 PM EDT spelling-context update closed a separate visual mismatch by carrying the source key signature into the transcription panel and requiring visual-range and score-spelling agreement. The 2026-05-14 10:52 PM EDT visual-crop demotion then rejected the same source crop because Alan's review showed that the visible source-score notes still did not match the transcription. Generated notation remains available only as the transcription-side rendering; a score panel must use a non-generated source-score image whose visible noteheads, register, spelling context, parsed score-note range, and explicit visible-note-sequence verification all match the transcription. The 2026-05-15 05:00 PM EDT strict evidence-gate update makes that demotion enforceable in code: a score crop cannot count as visible accepted evidence unless score box centers, visible score notes, written spelling, exact note-plus-octave sequence, audio-to-transcription agreement, transcription-to-score agreement, and an accepted truth-evidence flag all pass. The 2026-05-14 02:51 PM EDT target-triage update checks four-note-or-longer reference-audio targets against the current verified symbolic score excerpt before any promotion attempt. In that strict-gate state, the top source target was D D D# D D A# G; under the latest corrected source map its best exact-MIDI overlap is the single G5, so it remains not accepted score evidence. The 2026-05-14 03:49 PM EDT score-map workbench update added a local source-map verification queue with 779 unaccepted score glyph candidates across 7 staves from page 2 of the local IMSLP solo part. The 2026-05-14 04:24 PM EDT note-hypothesis update filters that queue into 172 unaccepted likely notehead hypotheses with staff-position pitch labels. The 2026-05-14 05:17 PM EDT review-packet update fixed the skipped-staff failure mode in the source detector, expands page-2 coverage to 951 unaccepted score glyphs and 214 unaccepted note hypotheses across 8 staves, and writes 8 staff-level source review packets. The 2026-05-15 12:45 AM EDT truth-workbench update adds a persistent accepted/rejected/pending truth-item path and a compact site surface for queued source targets, accepted truth clips, rejected mismatches, and regression benchmarks; it also makes the visible-note-sequence flag mandatory for score-visual and actual-source score locks. Before the strict-source correction, the deployed strict-gate tracker reported 43.126% exact

weighted progress, with 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, 0 score-visual locks, 0 actual-source score locks, 0 accepted truth items, 2 source-verification targets queued for truth review, 14 pitch-sequence groups, 2 source-target score checks, 0 verified source targets, 7 verified symbolic score notes, 0 verified source snippets, 951 unaccepted score glyph candidates, 214 unaccepted score-note hypotheses, and 8 staff review packets. That was implementation progress only; at that point a five-note candidate, rejected source-target checks, queued truth checks, and unaccepted score-glyph, note-hypothesis, or review-packet candidates did not solve full long-phrase or full-session transcription.

The 2026-05-15 06:34 PM EDT strict-source correction temporarily promoted one narrow 2026-05-03 Scherzo-Tarantelle lane: detected notes A#4 D5 C5 A#4 D5, displayed as Bb4 D5 C5 Bb4 D5, matched to an octave-aware score sequence, and backed by a non-generated source-score crop from the local Scherzo-Tarantelle solo part. Alan's later review rejected that five-note phrase as still incorrect. Curtis now treats the 06:34 PM state as a superseded attempt, not accepted evidence. The current strict state has 0 accepted score-linked phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, and 0 verified source snippets from that lane.

The 2026-05-15 08:37 PM EDT fast-transition candidate update addresses the next technical blocker directly. The broader 2026-05-03 Scherzo-Tarantelle audio stream is still not trustworthy as notation: the pYIN/onset path sees many repeated D/B-flat/G events and misses enough fast motion that no longer exact score phrase can be promoted without faking evidence. Curtis now stores a hidden YIN frame-transition trace as `scoreMatchCandidateNotes`. These candidates are used only as search input for the strict score matcher. They do not render as accepted notation, do not change practice time, and cannot promote a score snippet unless the same ordered detected MIDI notes match parsed source-score MIDI notes, the written score spelling and crop agree, local audio/video are present, and no matched detected note is marked uncertain by octave correction. A local check on the 5/3 Njh8_zq9_DM-9465 clip produced 174 hidden candidate notes with 8 pitch classes. After the later five-note rejection, those candidates remain search data only; no visible result is accepted.

The 2026-05-15 08:48 PM EDT notation-renderer correction moves the SVG treble clef baseline upward and re-anchors flat key-signature glyphs so B-flat and E-flat appear on the intended staff positions. This is a display correction only; it does not promote any transcription or score match without the stricter audio/source-score gate.

The 2026-05-15 09:07 PM EDT targeted-transcription update adds a direct reprocess path for specific

media samples through `/api/curtis/transcribe/run?sampleId=...`. This reduces the burden of waiting through broad eight-sample queue passes when a known evidence clip needs to be regenerated under a new transcription pipeline. It is an implementation-speed improvement, not an acceptance shortcut: targeted runs still use the same strict hidden-candidate and source-score gates.

The 2026-05-15 09:20 PM EDT live targeted-reprocess verification used that route on Njh8_zq9_DM-9465. The deployed API then reported 67.375% exact weighted progress, 1 accepted long-phrase counter item, 1 accepted measure-match counter item, 1 exact-source visible score group for 2026-05-03, and 1380 hidden transition-trace candidates across the stored transcription results. That public state is now superseded: Alan reviewed the displayed Bb4 D5 C5 Bb4 D5 phrase and rejected it. The useful retained result is the targeted-reprocess route and hidden-candidate search path, not the promoted phrase.

The 2026-05-15 09:24 PM EDT post-deploy check showed the live scanner had processed additional stored transcription samples after deployment: the live API reported 68.125% exact weighted progress and 2476 hidden transition-trace candidates. That visible one-group state is now rejected rather than accepted. The current correction requires the live counter to return to 0 accepted long phrases and 0 accepted measure matches until a new phrase survives review.

The 2026-05-15 10:05 PM EDT five-note rejection lock is the current acceptance state. Alan rejected the displayed five-note Scherzo-Tarantelle phrase itself, so Curtis now demotes A#4 D5 C5 A#4 D5 / Bb4 D5 C5 Bb4 D5 from accepted evidence back to candidate-only material. Regression tests require this exact lane to have zero verified score location, zero verified source snippet, no accepted measure, no accepted long phrase, no score-coordinate heat-map fragment, and no visible accepted score panel. The only note detections Alan had described as useful remain short audio-only D/A detections; the A4/A5 score anchors, D–B-flat–G–D score sequence, and five-note Scherzo phrase are rejected until new independent review proves otherwise.

The 2026-05-15 10:54 PM EDT second-pass audio-gate update targets the loop that caused repeated false progress. The hidden fast-transition trace could generate plausible note sequences before each note had independent audio confirmation, and reference-audio pitch traces could create short overlaps that looked like matches without being usable phrase evidence. A local audit of the rejected 2026-05-03 Njh8_zq9_DM-9465 clip read the candidate closer to Bb4 D5 Bb4 D5; the displayed middle C5 did not survive the second pass. Curtis now preserves audio-agreement provenance in daily detected-note series and filters hidden `scoreMatchCandidateNotes` before score search unless every note has

an independent agreement source, spectral/onset participation, adequate confidence, and adequate duration. Symbolic score matches and reference-sequence matches now use the same per-note audio gate. Reference-audio pitch-trace matches require at least four notes and three distinct pitch classes, so one- or two-note coincidences do not count as phrase progress. Regression tests cover the rejected Scherzo phrase with a weak C5, unagreed fast candidate notes, uncertain octave-corrected candidates, and weak reference-audio matches. This update does not create an accepted long phrase; it removes a false-positive lane before the next phrase attempt.

The 2026-05-15 11:17 PM EDT candidate-isolation update closes the remaining daily-record leak. Source-verification candidates and reference-audio pitch-sequence groups are now separated from displayable score matches. Candidates may populate `candidateMatchGroups`, the truth workbench, and source-verification target queues, but the user-facing `matchGroups` bucket stays empty until exact source-score crop, visible note sequence, box centers, paired audio, transcription, score spelling, and accepted truth-evidence flags all pass. A regression test covers a playable Wieniawski source-sequence candidate that remains queued without becoming a visible match, score link, reference link, or heat-map fragment.

The 2026-05-15 11:32 PM EDT live verification reported the strict post-update state: 46.125% exact weighted implementation progress, 0 accepted long phrases, 0 accepted measures, 0 exact score-aligned windows, 0 actual-source score snippets, and 0 score-coordinate heat-map fragments. The 2026-05-03 daily record had 0 visible `matchGroups`, 6 queued `candidateMatchGroups`, and no score or reference link. The current top queued target is A D D F# D, which remains rejected as accepted score evidence because it does not form a contiguous exact-MIDI run in the corrected source map.

The 2026-05-16 source-position and exact-MIDI update closes the next source-map error. Re-rendering the local IMSLP solo part at high resolution and cross-checking the Wieniawski Society bars 5–9 example showed that the opening violin entrance is A5 G5 F5 A5 G5 F5 A5 G#5 F5, not the prior D6 C6 Bb5 D6 C6 Bb5 D6 C#6 B5 map. The source MusicXML now stores those nine actual source notes from measures 5–7, and the older lower-octave and wrong-staff-position maps are rejected by regression tests. Curtis also removes the remaining pitch-letter-only loophole from source-verification targets. A longer target may still enter the queue from pitch-class search, but the score check now fails closed unless the detected notes carry exact MIDI values, every detected note passes paired-audio agreement, and the detected MIDI sequence appears contiguously in the parsed

source MusicXML MIDI sequence. The new positive regression target is G5 F5 A5 G#5 F5, whose
 detected MIDI sequence 79 77 81 80 77 matches the same contiguous source MIDI sequence
 exactly. This does not promote any current live Scherzo-Tarantelle phrase; it prevents A matching a
 wrong score note, a wrong register, or a staff-position mistake from being reported as score evidence.
 The 2026-05-16 exact-MIDI truth-manifest update turns that rule into a reusable source of truth
 instead of leaving it as scattered scanner logic. Curtis now carries a long-phrase truth manifest that
 verifies one source-only positive phrase, G5 F5 A5 G#5 F5, against the Wieniawski MusicXML and
 blocks three known false phrase families: the old D6 C6 Bb5 opening map, the user-rejected Bb4
 D5 C5 Bb4 D5 lane, and the current queued G5 G5 A#5 A#5 A#5 A#5 A5 target. A new shared
 gate computes the exact MIDI sequence, the best contiguous source overlap, the ordered overlap, and
 the audio-agreement state before any source target can be considered verified. The site now exposes
 source-truth and blocked-error counts in the compact solve-status panel. At that point this advanced
 the implementation infrastructure only: accepted live phrase count remained zero until a paired local
 audio/video phrase passed the same exact-MIDI and audio gate.

The 2026-05-17 Staff 4 source-lane extension deliberately grows the current accepted source lane
 instead of starting another unrelated match. The verified MusicXML now extends the accepted Eb5
 Eb5 C5 Eb5 Eb5 Staff 4 window with the two immediate right-adjacent source notes, giving a source
 sequence Eb5 Eb5 C5 Eb5 Eb5 Eb5 C5. A new actual-score crop from the same local IMSLP page
 shows the right-1 six-note subrange with centered boxes, and the current May 3 audio run D#5 D#5
 C5 D#5 D#5 D#5 resolves to the identical MIDI sequence 75 75 72 75 75 75. Curtis therefore
 promotes the six-note right-1 subrange as accepted source/audio truth. The seven-note right-2 lane
 remains source/candidate material until its seventh note, source crop range, local audio/video playback,
 and truth-review state pass the same gate.

The 2026-05-17 anchored-expansion harness makes that source-lane discipline reusable. Instead of
 looking for a new unrelated score match after one accepted phrase exists, Curtis now enumerates
 outward ranges from the accepted Staff 4 anchor, tests left and right adjacent source-note windows
 against available audio-agreed note runs, and records the exact blocking note when the expansion
 fails. In the current state the harness promotes the right-1 six-note lane Eb5 Eb5 C5 Eb5 Eb5 Eb5
 because its exact source MIDI, audio-derived MIDI, actual source crop, and truth manifest agree. The
 next candidate is the right-2 seven-note lane Eb5 Eb5 C5 Eb5 Eb5 Eb5 C5; it must be audited as a
 new adjacent window instead of inheriting acceptance from the six-note prefix. This is implementation

progress because the next step is now mechanically constrained: keep the same source lane fixed and accept or reject the seventh note under the same gate, rather than restarting with a random candidate.

The next 2026-05-17 expansion-search update broadens that fixed-lane search without weakening the evidence gate. The harness no longer searches only the already-ranked score-candidate cards, because those cards can hide the exact adjacent audio run needed to extend a verified phrase. It now also mines the raw detected note series stored inside each daily record, preserves whether a run came from a ranked match group or raw detected-series pool, and reports the raw-run count in the completion tracker. This is deliberately not an acceptance shortcut: a raw run can only become visible evidence if its exact MIDI sequence, paired-audio agreement, actual source crop, and accepted truth state all pass. If the raw pool does not contain the next Eb5, the tracker still blocks the lane and reports the searched audio-run count instead of starting a new random match.

The 2026-05-17 03:23 AM EDT source-audio rescanner extends that search beyond the currently stored note windows. Curtis now starts from the accepted Staff 4 source/audio anchor, locates the original local media sample, extracts a padded source-audio window around the accepted clip, reruns the current note detector on that fresh audio, shifts the detected notes back into the source-video timeline, and adds those runs to both the anchored-expansion harness and the adjacent-mining pass. The rescan cache is keyed by media path, size, modification time, source window, and transcription-pipeline version, so source-media changes or detector changes invalidate stale runs. This adds search coverage, not acceptance. A source-rescanned run can only become a phrase candidate if its exact MIDI sequence matches the verified Staff 4 source lane; it still needs paired-audio review, source crop agreement, local media playback, and truth acceptance before it can display as a score match.

The 2026-05-17 Staff 4 anchor-persistence update fixes a state-regression risk in that search path. The accepted Staff 4 anchor is now recovered from the long-phrase truth manifest when the current daily record has zero visible accepted `matchGroups`. That manifest entry carries the source id, exact MIDI sequence 75 75 72 75 75, practice day, source media sample, source window, and local timing for the accepted audio phrase. The source-audio rescanner, anchored-expansion harness, and adjacent-mining pass all consume this persistent anchor source and deduplicate it against any visible accepted match-group anchor. The live site therefore cannot fall back to `no_staff4_anchor` merely because the display layer has no promoted match card; it must either rescan the verified Staff 4 source sample, report missing local media, or report the next exact-MIDI search result. This still does not accept a longer phrase. It preserves the one verified anchor so future phrase work starts from the same

source/audio truth instead of restarting with unrelated candidates. 233

The 2026-05-17 12:56 PM EDT tiled-rescan update widens the next source-audio search without 234
changing the acceptance rule. The Staff 4 source-audio rescanner now plans up to five overlapping 235
source-window tiles from the same accepted anchor instead of one short padded clip: the anchor-core 236
tile plus four rightward tiles that can cover the rest of the *8835-8925 source sample. Each tile is 237
extracted, transcribed, shifted back into the original source-window timeline, cached under a versioned 238
key, and merged into the same exact-MIDI expansion and adjacent-mining pools. This is search 239
coverage only. A tiled run remains candidate data until the source MIDI, audio-derived MIDI, paired 240
audio, source crop, local media, and truth review all agree. 241

The 2026-05-17 01:09 PM EDT anchor-reproduction check closes a loophole in that widened scan. 242
The deployed tiled pass processed five source-window tiles and produced four note runs with 286 243
primary-plus-candidate note events, but adjacent mining still found zero exact Staff 4 right-extension 244
candidates after 1547 stored/rescanned MIDI windows. More importantly, the fresh current-detector 245
rescan did not reproduce the already accepted five-note Staff 4 anchor 75 75 72 75 75. Curtis now 246
records that as an explicit `not_reproduced` anchor state. The next step is therefore not another blind 247
widening of the same search window; it is detector/segmentation calibration or a newly accepted 248
current-detector anchor before any longer Staff 4 phrase can be promoted. 249

The 2026-05-17 01:43 PM EDT anchor-guided reproduction update fixes that current-detector 250
regression without weakening the evidence gate. The source-audio rescanner now uses the accepted 251
Staff 4 note windows as timing seeds, rechecks each exact audio segment with the current pYIN, YIN, 252
and spectral-onset detectors, and adds a reproduced anchor run only when the detector votes agree on 253
the exact MIDI values 75 75 72 75 75. A broad tiled transcription can still miss the anchor, but it 254
can no longer force the tracker back to `not_reproduced` when the current detectors independently 255
confirm those accepted windows. Live verification after deployment reported five rescan windows, 256
five source-audio runs, 291 primary-plus-candidate-plus-guided events, five guided anchor events, and 257
`reproduced` anchor status; adjacent mining still found zero exact right-extension candidates after 258
1671 stored/rescanned MIDI windows. The same pass also fixes transcription rendering for accidentals: 259
a detected A# can display as Bb in a flat-key context or with a local accidental, so the staff notehead 260
no longer silently appears as natural A. A 2026-05-17 02:19 PM EDT renderer patch makes that gate 261
visible rather than font-dependent: local sharps and flats are now drawn as SVG accidentals, high-note 262
accidentals are clamped inside the staff viewport, and Bb/Eb key signatures are rendered from the same 263

flat glyph path used for local flats.

The 2026-05-17 04:49 PM EDT adjacent-guided phrase pass implements the next segmentation step after that anchor was recoverable. The rescanner now derives the immediate right-side Staff 4 source targets from the accepted lane itself, currently 75 75 72 75 75 75 and 75 75 72 75 75 75 72, infers adjacent note windows from the accepted anchor timing, and probes each expected MIDI value with the same current pYIN, YIN, and spectral-onset vote gate used for anchor reproduction. Curtis only adds an adjacent phrase run if every note in the source target is reproduced by exact current-detector MIDI agreement; otherwise the record stores the failed note index, expected MIDI, detector votes, and seed window rather than showing a match. This does not solve long-phrase transcription by itself. It removes another source of circular work: the system can now distinguish “the broad transcriber missed it” from “the current detectors cannot reproduce the next exact source note at the inferred window,” which is the precise blocker that must be fixed before the Staff 4 lane can grow beyond the accepted six-note phrase.

The 2026-05-17 05:29 PM EDT adjacent-onset sweep update tightens that same gate. The inferred Staff 4 continuation no longer gets only one guessed next-note window. For notes beyond the accepted anchor, Curtis now tests a bounded early/late onset sweep around the inferred start time, keeps the accepted anchor windows fixed, and records each attempted offset, detector vote set, exact-vote count, and best failed attempt. If a nearby offset reproduces the expected MIDI with pYIN, YIN, and spectral-onset agreement, the note is accepted with its actual timing offset; if every offset fails, the phrase remains blocked with a precise detector-audit trace. This step targets the likely failure mode where the sixth source Eb5 exists in the audio but the inferred window lands slightly before or after the played onset.

The 2026-05-17 05:54 PM EDT failure-localization and notation-quality pass turns that detector trace into a usable next target. The source-audio rescanner now extracts the first failed adjacent Staff 4 note from all guided targets, preserving the target direction, target MIDI sequence, failed note index, expected note, best swept offset, detector votes, and the detector consensus note heard at that offset. The completion API and compact site tracker expose that single blocker instead of a broad “not reproduced” label, so the next calibration step can distinguish a missing detector vote from a wrong-MIDI segment such as source Eb5 with detected D5. The same pass raises the notation-rendering gate: sharp glyphs now use narrow stems and filled slanted bars, flat glyphs use a separate curved stem and filled bowl, key-signature flats share the same engraved flat path, and the treble clef baseline is adjusted so the

G loop is aligned with the G line. Regression tests now cover accidental structure, clef placement, key-signature flats, high-note accidental clipping, and preservation of note labels, because correct transcription data still fails the handoff if the rendered notation looks amateur or misleading. A 2026-05-17 06:14 PM EDT visual QA correction then fixed the remaining clef clipping by giving all notation and notation-gate SVGs a taller engraving viewport and by preventing compact matched panels and responsive overrides from shrinking or restyling clefs, sharps, and flats into a lower-quality mobile rendering.

The 2026-05-17 08:40 PM EDT Staff 4 first-failure audit gate makes the next phrase decision explicit instead of letting a candidate window blur into evidence. The packet generator now prefers the written source spelling from the Staff 4 target sequence, so the score-side failed note stays Eb5 while detector spelling such as D#5 remains a separate diagnostic field. Each packet now records a decision object: accepted only when exact audio MIDI and source-score truth are both locked, rejected when detector consensus hears a different MIDI value, and queued for gold review or rescan when current detectors do not reproduce the source note. The visible site reads that decision rather than just the raw packet status. This prevents the current lane from extending on a stale stored run, an unverified source crop, or a sharp/flat spelling artifact.

The 2026-05-17 10:46 PM EDT Staff 4 right-1 promotion applies that audit gate to the first adjacent note. The accepted source side is Eb5 Eb5 C5 Eb5 Eb5 Eb5; the detected audio side is D#5 D#5 C5 D#5 D#5 D#5; both reduce to exact MIDI 75 75 72 75 75 75. The new non-generated source crop `wieniawski-scherzo-tarantelle-staff4-eb-eb-c-eb-eb-eb-verified.png` boxes those six noteheads from the actual IMSLP solo-violin page, and the truth manifest now contains two live accepted Staff 4 source/audio phrases and four verified positive source phrases. This is still a measure-window, not solved long-phrase transcription. The right-2 seven-note lane remains the next adjacent audit target; it cannot display as accepted until its full source sequence, transcription, local audio/video, crop range, and truth review agree.

The 2026-05-16 09:07 PM EDT gold-review-loop update adds the missing human acceptance layer between candidate extraction and training data. Curtis now queues reviewable audio-phrase and score-phrase candidates from detected local media windows, exposes them through `/api/curtis/gold-review`, and writes accepted, rejected, or pending labels through `/api/curtis/gold-review/items`. Accepted audio phrases become gold audio truth only; they do not become score evidence. Accepted score phrases must carry local media, exact accepted note names

with octaves, exact score note names with octaves, a score location, and a source score coordinate before they can become score-ready truth. The same write path mirrors accepted and rejected labels into the existing truth workbench, so the correction loop, implementation percentage, regression tests, and compact site surface all share the same source of truth. The 2026-05-16 09:36 PM EDT playable-window guard then tightened the review queue: a gold candidate is not published unless its local audio clip is short enough for the site playback endpoint and carries a real local media route. This update still does not solve long-phrase transcription; it creates the review loop needed to stop candidate snippets from becoming public “matches” or dead review rows before they are checked.

2.1 Media indexing

The source inventory stores platform, title, publication time, duration, view count, candidate reasons, media kind, and blocker state. Dated long-form practice logs are treated as high-value longitudinal records because the title contains a practice date and the duration preserves session-level evidence. The uploaded-video ledger starts at the `violin 1` marker and counts only violin-numbered or date-titled practice logs, so unrelated long public videos are retained in inventory without being silently counted as violin practice. This ledger is not the active-practice total. Active practice is measured only after media processing identifies windows where violin playing is actually occurring.

2.2 Review evidence

The review layer stores section-level findings with a dimension, evidence source, judgment, and practice constraint. The current dimensions include tone, intonation, shifts, time, articulation, and audition delivery. Findings can be strong, needing work, or unjudged. The unjudged state is part of the system design: if the sound is not available, the hand is obscured, the camera angle hides a transition, or still frames do not support a timing claim, the system should not convert weak evidence into a confident critique.

Table 1. Curtis source state checked at 2026-05-15 05:30 PM EDT.

Signal	Current value
Primary source	YouTube channel @nalalan
Inventory	219 indexed videos
Practice candidates	112 videos
Long-form candidates	92 videos
Uploaded-video marker	violin 1, uploaded 20 December 2025
Uploaded-video ledger	48 videos, 234 h 57 min uploaded session duration
Estimated total practice time	223 h 5 min extrapolated from the checked-window active-playing ratio; provisional because the checked windows remain sparse and sample-biased
Measured practice time	2 h 21 min of detected violin-playing footage; this is independent from transcription and cannot exceed checked video coverage
Video checked for practice time	2 h 28 min of the 234 h 57 min uploaded-video ledger
Active-practice scan	331 active sub-intervals persisted from stored media samples; 54 min 2 s of scan-derived active audio; 106 sample results; violin 1 windows through *2160-2250 checked; 250 queued windows still pending owner media
Unchecked archive duration	232 h 28 min still awaiting violin-playing detection
Media samples	106 active-scan sample results; 94 active violin; 12 checked non-violin
Reviewed sections	80 sections
Curtis-focused findings	80 findings
Latest dated practice log	5-6-26
Review model	GPT-5.5 for vision/text, separate audio models for audio evidence and piece verification
Confirmed labels	5/1 Haydn Symphony No. 94 IV; 5/2 and 5/3 Wieniawski Scherzo-Tarantelle
Training sources	3 source-confirmed practice sources
Score-reference targets	3 configured
Reference training	3 source-confirmed score targets, 22 pitch/rhythm windows, title-labeled calibration sources for future scale/arpeggio uploads, and 4 official YouTube Data API public-reference queries kept separate from repertoire claims and audio evidence
Analyzed practice days	41 indexed days; 3 practice-time measured daily records; 3 audio-evidence records; 2 score-audio-only records; 1 detected-note-fragment record; 3 confirmed source labels, score targets, and a source-backed Wieniawski nine-note symbolic opening phrase now corrected to A5 G5 F5 A5 G5 F5 A5 G#5 F5
Machine pitch events	Detected-active-section onset-aware pYIN plus onset-bounded spectral pitch rescue for fast passages, audio-agreement counts, pitch-sanity filtering, repeated-pitch-collapse guarding, practice-coverage fields, title-labeled calibration handling, and fingerprints stored only when quality gates pass; broad candidate micro-transcription is withheld, while detected note series are preserved in the transcription PDF and promoted to note-match groups only after pitch-class sequence agreement with an available score-derived or reference-audio sequence
Specific observations	Rendered only when grounded in reliable score-linked evidence or repeated score-free technique-pattern evidence
Note-match windows	1 accepted score-linked six-note Staff 4 window, extending the earlier accepted five-note prefix; the rejected Bb4 D5 C5 Bb4 D5 Scherzo-Tarantelle lane remains blocked; D/A audio-only detections remain useful note evidence but are not score-location matches
Transcription solve tracker	Current state after the Staff 4 tiled-rescan, anchor-guided reproduction, adjacent-guided phrase pass, adjacent-onset sweep, first-failure localization, notation viewport QA, and right-1 audit promotion: 1 accepted exact-MIDI source/audio measure-window is visible for the six-note Staff 4 lane, still not a solved long phrase; the accepted Staff 4 anchor is recoverable from the truth manifest even when the daily display has zero visible accepted match groups, and the current detector now has seeded reproduction paths for both the accepted anchor and its immediate right-side source targets; source-audio search uses up to five overlapping tiles from the same *8835-8925 source sample; the accepted right-1 lane stores source Eb5 Eb5 C5 Eb5 Eb5 Eb5, detected audio D#5 D#5 C5 D#5 D#5 D#5, exact MIDI 75 75 72 75 75 75, local timing 20.225-22.992, and an actual-source crop with centered six-note boxes; the right-2 seven-note lane is still candidate/source material until its seventh note and full crop/audio/truth state pass review; expansion search includes raw detected note series, ranked candidate cards, truth-manifest anchors, source-audio-rescanned runs, and exact adjacent-guided runs only when every note passes pYIN/YIN/spectral exact-MIDI agreement; a Staff 4 phrase-audit packet generator can extract the next audio/video window, rerun pYIN/YIN/spectral checks, and store pitch-trace and spectrogram artifacts without accepting a failed expansion; notation rendering now has regression and visual coverage for clef baseline, unclipped clef viewport, sharp/flat glyph structure, key-signature flats, high-note accidental clipping, responsive symbol styling, and accidental labels; 4 source-positive exact-MIDI phrases are verified against MusicXML; 4 known false phrase regressions are blocked; 2 live Staff 4 source/audio truth phrases are accepted in the manifest; gold review candidates can now be accepted or rejected before training; remaining score-glyph review packets stay implementation support, not accepted evidence
Daily evidence surface	Local sample playback only for violin-positive windows, notation withheld until reliable note/rhythm verification, score snippets hidden until played-to-score agreement, and explicit whole-session transcription limit
Curtis-level completion	Not scored when the available evidence only identifies repertoire
Current focus	Capture a playable audition-view take
Current constraint	Record 30 seconds of sustained tone
Boundary	Curtis admission cannot be predicted from current samples

3 First readout

The first live readout is experimental and intentionally conservative. The strongest immediate value is that the system has already converted a large, timestamped practice archive into a queryable corpus with captured media samples, stored section reviews, and title-confirmed practice records. The home screen has three top-level surfaces: a paper card, analyzed practice days, and repertoire, followed by a compact transcription-completion tracker and gold review surface at the bottom. The first evidence row now reports total practice time first, then the video already checked for violin-playing, then the 234 h 57 min YouTube practice archive; this prevents the public practice ledger from being confused with either transcription coverage or uploaded-video duration. Each analyzed day groups same-day videos, preserves uploaded duration, and names confirmed or uncertain piece evidence without promoting guesses. Playable local media samples, paired audio/video evidence windows, and practice-time credit are now withheld unless violin playing is detected in the source window. The owner-sync sampler now reuses locally cached captured media, uploads cached violin-positive windows before blind new captures, resumes interrupted queue-only runs from cached pending-window media including non-playing windows, consumes active-scan pending windows before generic expansion or blind sampling, has a queue-only mode for chronological pending-window coverage, expands around already-found violin-positive windows, and can switch into full-check mode to walk every 90-second window in order instead of staying limited to sampled windows; the live analyzed-days surface currently exposes violin-playing practice-time evidence for 5/1, 5/2, and 5/3, and the first twenty-five violin 1 windows are now represented as checked coverage, with three non-playing/unclear windows and twenty-two active windows. Broad sheet-music notation is also withheld until reliable note/rhythm verification, so audio evidence cannot appear as a failed transcription. The current visible note evidence includes exact audio-checked D4 and A4 fragments; the five-note Scherzo-Tarantelle phrase from 5/3 is rejected and stays candidate-only. Single-note Scherzo-Tarantelle score crops remain rejected. A multi-note score panel now renders only when the detected MIDI sequence, the parsed score MIDI sequence, the written spelling context, a non-generated source-score crop, and musician review agree on the same visible range; generated notation SVGs are no longer allowed to stand in for the score panel. A source-confirmed piece title creates a score target, and a detected pitch-class sequence match can show the played clip and transcription, but broad score-region estimates and visually unverified crops do not render as sheet-music snippets. The gold review surface can queue detected audio phrases and possible score phrases for acceptance or rejection; accepting an audio-only label does not promote it to

score evidence, and accepting a score label requires exact audio/score note-and-octave agreement plus
 score location. The daily record includes a compact source-label field so a missing or wrong day title
 can be corrected without letting that correction masquerade as score alignment. The analyzed-days
 surface no longer promotes broad candidate transcription while candidate micro-transcription fails
 audible match review; those windows remain hidden support data or audio evidence only. Active scans
 with no confirmed piece are treated as possible repertoire or score-free technique exercise evidence
 rather than as failed score matching. The repertoire surface is evidence-backed: an entry appears only
 when a daily record carries confirmed source evidence, and each entry keeps practice dates, clips,
 score targets, practice time when measured, and current blocker limits. Generic etude/caprice evidence
 remains outside the piece-title field, and possible or uncorroborated repertoire labels remain checkable
 against the source window. These statements are not generic pedagogy; they are extracted from stored
 review findings, sample indexes, source corrections, public-domain score targets, pitch-quality checks,
 accepted/rejected gold labels, and the uploaded-video ledger.

After the 2026-05-15 10:05 PM EDT five-note rejection lock, the previously shown Bb4 D5 C5 Bb4
 D5 group is blocked from accepted display. The 2026-05-16 10:30 PM EDT Staff 4 source-review
 pass then added the first accepted score/audio measure-window lane: the local IMSLP source crop
 shows Eb5 Eb5 C5 Eb5 Eb5, the paired May 3 detected phrase stores D#5 D#5 C5 D#5 D#5, and
 both resolve to the same exact MIDI sequence 75 75 72 75 75. The 2026-05-17 10:46 PM EDT
 right-1 promotion extends that lane by one accepted note: the source crop shows Eb5 Eb5 C5 Eb5
 Eb5 Eb5, the paired May 3 detected phrase stores D#5 D#5 C5 D#5 D#5 D#5, and both resolve to
 exact MIDI 75 75 72 75 75 75. The right-2 seven-note lane is still not accepted. The Staff 4 audit
 packet generator now turns the next adjacent decision into an inspectable packet: source target, stored
 audio run, paired local audio/video window, pYIN/YIN/spectral pitch checks, onset list, pitch trace,
 spectrogram, and a truth decision. The acceptance rule remains severe: if a source crop, note label,
 transcription glyph, audio-derived MIDI sequence, or musician review disagrees, the group must drop
 back to hidden candidate data instead of appearing as a match.

Current transcription limit. Machine note output now has four display gates. First, the media
 window must be classified as violin-positive before it can appear as playable evidence or count toward
 practice time. Second, the note stream must audibly match the paired audio before any notation
 can render. Third, candidate note series must pass the gold review loop before they can train or

Table 2. Current evidence classes and limits.

Class	Stored signal	Limit
Audio	Tone, intonation, shifts, time, articulation	Some audio reviews fail or cannot inspect the excerpt cleanly
Video	Setup, posture, bow lane, visible hand/bow evidence	Still frames can hide motion, sound, and transition accuracy
Metadata	Dates, duration, channel, view count, practice-candidate labels	Metadata cannot judge playing quality by itself
Progress plan	One focus, practice constraint, short session plan	The plan is a next experiment, not an outcome prediction

count as accepted truth. Fourth, repertoire passages need score alignment, while score-free exercises need repeated-pattern grouping before they can become full practice transcription. Source-confirmed score targets are also display-gated: broad score snippets are hidden unless the corresponding played window has explicit score-location agreement, and single-pitch score crops require explicit visual note verification before they can render. Score-note sequences are now source-gated before matching: a scanned PDF plus guessed `scorePitchClassSequences` is ignored unless the note sequence has an explicit symbolic-score source, manual source confirmation, `MusicXML`/source-confirmed status, or exact score-location metadata. Curtis now includes a symbolic-score parser for inline `MusicXML`, local `.musicxml`, and local `.mxl` targets. That path extracts score notes, measures, pitch classes, written note names, octave-aware MIDI values, and duration fields. The parser now preserves written flats such as B-flat and E-flat for score staff placement while matching their enharmonic pitch class to detected audio, so a source-score flat can no longer be rendered on the wrong staff position as a sharp-only spelling. The Scherzo-Tarantelle target now carries a source-backed nine-note symbolic opening phrase from the local IMSLP solo part, one accepted Staff 4 source/audio window Eb5 Eb5 C5 Eb5 Eb5 Eb5, and one Staff 4 source-only extension Eb5 Eb5 C5 Eb5 Eb5 Eb5 C5. The accepted six-note window has a centered actual-source crop and exact MIDI agreement with the paired May 3 audio; the seven-note extension remains candidate/source material until its full adjacent note, crop range, local media, and truth review agree. The audit generator can produce a bounded evidence packet for the next adjacent decision, and the source-audio rescanner can reopen the original Staff 4 media around the accepted anchor to search beyond the previously stored note windows. Neither path accepts evidence by itself: audit packets document a blocker, and rescanned runs remain candidate search data until exact MIDI, paired audio, source crop, and truth review agree. The latest reviewed five-note crop for Bb4 D5 C5 Bb4 D5 remains rejected and blocked from accepted score display. A

visible score match normally requires at least five contiguous played notes, exact note-and-octave
MIDI agreement, paired-audio agreement, and an actual source crop; the Staff 4 lane proves that a
narrow measure-window can pass only when those exact source and audio gates all agree. One-note
overlaps and repeated single-pitch runs are still rejected as score evidence. Hidden fast-transition
candidates and reference-audio pitch traces now also require second-pass audio agreement for every
matched note before they can create a candidate phrase; weak one- or two-note overlaps are withheld
as search data, not progress. When the symbolic gate passes, generated notation can render candidate
transcription support, but the score panel is accepted only if it uses an actual source-score crop
whose visible noteheads, register, written spelling, exact range, paired audio, transcription, and
musician review match; a generated SVG excerpt can no longer masquerade as the actual score.
Source-verification targets now require exact detected MIDI values and per-note audio agreement
before even the source-check status can become a score-sequence verdict; pitch-class overlap alone
is recorded only as search context. All prior Scherzo-Tarantelle single-pitch crops and the rejected
5/3 five-note crop remain withheld because review found that the visible score evidence did not
match the displayed transcription. These failures are now regression fixtures. Unverified room, setup,
laptop, silence, or non-playing samples remain withheld, and unverified pitch traces remain hidden
support data rather than visible notation. A spectral-onset rescue path estimates pitch inside short
onset-bounded violin segments when the pYIN path collapses, while the repeated-note trace guard
prevents fast or complex playing from becoming a dominant repeated pitch, records the quality mode,
and keeps unusable traces out of accepted score evidence. The current micro-transcription path stores
candidate short contiguous runs, but it does not render them as sheet music or count them as solved
transcription until the displayed notes pass paired-audio listening review and, for score evidence, exact
audio/score note-and-octave review. The transcription-run PDF preserves all stored detected note
series for the day. Source-score promotion uses exact MIDI sequence agreement plus paired-audio
agreement; looser pitch-class grouping remains candidate-only and reference-audio matches alone
are not treated as score notation. A separate audio-checked fragment gate can render a note when
pYIN and YIN agree on the same semitone over a short stable window; the earlier live examples were
D4 and A4, and the new accepted source/audio lane is still only a six-note measure-window rather
than a long phrase. Full-session transcription remains incomplete until every playing section in the
long-form practice videos is extracted, verified as violin-positive, verified against the audio, accepted
or corrected through review, and aligned either to a score section or to a repeated technique pattern.

The current rejection proves the strict lane is still not strong enough to solve long phrases. Curtis can now keep false score evidence out of the accepted surface, but it still needs a newly verified phrase that carries many consecutive notes through source-score parsing, actual-source crop selection, local audio/video pairing, transcription rendering, musician review, and regression tests without showing a wrong-note score crop. The next unsolved step is length: the same gate must work for many consecutive notes, then measures, repeated attempts, rhythm, and full active-playing windows.

4 Discussion

Curtis Media Review is intentionally conservative. It does not rank the player against applicants, infer admissions probability, or convert every video into a claim. Instead, it keeps a running technical record. The sampler now treats violin presence as a hard evidence boundary: playable clips and practice-time credit appear only after violin playing is detected in a media window, while notation waits for reliable note/rhythm verification and either score alignment or score-free pattern grouping. This makes the system useful even before it becomes musically sophisticated: it can preserve practice volume, surface recurring constraints, identify missing evidence, and make subsequent recordings easier to compare against earlier ones.

The main open problem is repertoire identification, score-free exercise representation, and evidence quality. YouTube metadata can identify candidate videos and public labeled reference sources, and the current owner-sync path can provide direct audio/video samples, but unclear camera framing, missing score/title context, improvised technical patterns, and short excerpts still prevent reliable piece naming. The backend therefore keeps generic labels out of the piece-title field, stores them as candidate evidence, requires source-window evidence for confirmed titles, and marks weak evidence explicitly. The current training ledger separates source-confirmed labels from title-labeled calibration uploads, public YouTube reference metadata, independent audio-title matches, score-free or piece-pending active windows, failed unverified machine pitch/rhythm windows, failed pitch-collapse windows, withheld candidate micro-transcription windows, audio-checked single-note fragments, detected note series, pitch-class reference matches, and true score-aligned windows. It also stores reference targets for the three confirmed source labels across two works so later passes can ask for section-level matching instead of repeating broad title guesses. The sampler classifies candidate windows using active audio, violin-range voiced pitch, onset density, pitch-class diversity, spectral features, and

minimum practice-duration gates before a sample can become visible evidence. The transcription path converts sampled media to mono audio, finds audio-active playing sections, runs pitch tracking only on those sections, normalizes and harmonic-isolates each section for pitch tracking, estimates monophonic violin pitch with an onset-aware pYIN pitch track bounded to the violin range, detects onset boundaries so repeated attacks on the same pitch are not silently merged into one sustained note, runs a second spectral pitch estimate inside short onset-bounded segments when fast playing defeats pYIN, compares pitch diversity across candidate note streams, checks whether selected note events agree with an independent spectral or pYIN/onset detector, rejects out-of-range pitch artifacts, requires sustained pitch changes for pitch-track segmentation, removes very short low-confidence pitch blips, marks isolated octave-flip corrections as uncertain, rejects repeated-pitch collapse, estimates tempo, stores detected note series for audit/PDF review, and builds pitch-class, interval, rhythm, and contour fingerprints only when quality gates pass. Daily records now carry transcription quality, material status, and practice-coverage states, and the primary interface withholds media playback unless the source window is violin-positive, withholds broad machine notation until note/rhythm verification, displays only audio-checked fragments when the stricter gate passes, hides loose reference-audio matches from the daily row, shows a source-score crop only when the highlighted score note has explicit visual note verification, and withholds broader score snippets until exact score-location alignment exists. A stronger system should align repertoire events against a symbolic or optical score representation, group score-free technical patterns by repeated note/rhythm fingerprints, identify movement sections or measure ranges when a score exists, compare against reference recordings available through permitted media, and then judge articulation, timing, tone, and tempo against the aligned passage or repeated pattern.

The broader experiment is whether a practice system can make musical development legible over time. A useful Curtis system should show whether tone, intonation, rhythm, articulation, shifts, memory, endurance, and audition delivery are improving across dated recordings. It should also preserve the source evidence behind each claim so that the dashboard does not become motivational text detached from the actual playing.

5 Roadmap to solved long-phrase transcription

This section is the working plan for making Curtis able to produce long, trustworthy audio–video–transcription–score evidence from Alan’s actual violin practice. It is intentionally part of the paper rather than a separate planning note. Whenever Curtis makes real progress toward full-session active-practice measurement, accurate transcription, score matching, heat maps, repertoire updates, or Curtis-level observation, this PDF should be updated with the new state, the acceptance evidence, and the remaining blocker.

5.1 Current tracker state

The tracker was refreshed through the 2026-05-17 10:46 PM EDT Staff 4 right-1 promotion. The system has a real media ledger, a deployed active-practice scan route, a working owner-sync path for captured media, a visible 100-point transcription-completion tracker with precise weighted progress, stored user-correction benchmarks, a persistent truth workbench for accepted/rejected/pending audio-score-transcription items, a local source-backed Wieniawski score PDF, a nine-note source-backed Wieniawski symbolic score opening phrase, a source-verification target queue, source-target score triage, an unaccepted score-glyph verification queue, an unaccepted score-note hypothesis queue, staff-level source review packets, a stricter evidence gate, a hidden YIN fast-transition candidate trace for score matching, a Staff 4 phrase audit packet generator, a Staff 4 adjacent-mining pass, a Staff 4 source-audio rescanner with five overlapping source-window tiles, current-detector anchor reproduction, and a truth-manifest anchor fallback that keeps the accepted Staff 4 lane available even when the daily display has no promoted matchGroups. The Wieniawski opening source map is A5 G5 F5 A5 G5 F5 A5 G#5 F5; the previous D6 C6 Bb5 D6 C6 Bb5 D6 C#6 B5 and lower-octave D5 C5 Bb4 D5 C5 Bb4 D5 maps are rejected and covered by regression tests. The 2026-05-03 Scherzo-Tarantelle five-note phrase A#4 D5 C5 A#4 D5, displayed as Bb4 D5 C5 Bb4 D5, is explicitly rejected after Alan’s review and cannot display as accepted score evidence. The old A4/A5 single-note anchors remain rejected, and a one-note pitch anchor still cannot render unless the score note and octave are explicitly verified. The accepted Staff 4 source/audio lane is now six notes: source spelling Eb5 Eb5 C5 Eb5 Eb5 Eb5, detected audio spelling D#5 D#5 C5 D#5 D#5 D#5, exact MIDI 75 75 72 75 75 75, local timing 20.225–22.992, and a centered actual-source crop from the local IMSLP solo-violin page. The right-2 seven-note lane remains candidate/source material until the full source sequence, transcription, local audio/video, source crop range, and truth review agree. The transition trace is not displayed as notation and fails closed unless every matched fast-note candidate has a second

detector agreement path, including spectral/onset agreement, sufficient confidence, and sufficient duration. Loose one- or two-note reference-audio coincidences no longer count as phrase progress, and candidate pitch-sequence groups no longer enter the daily matchGroups display bucket unless the exact source-score crop, visible note sequence, box centers, paired audio, transcription, score spelling, and accepted truth-evidence flags all pass. This is not solved long-phrase transcription yet; Curtis has one accepted six-note Staff 4 source/audio measure-window, but accepted long-phrase transcription remains blocked until a longer phrase survives audio, transcription, source-score, crop, and musician review.

Table 3. Progress tracker baseline for the long-phrase transcription goal.

Tracker item	Baseline state
Goal	Produce compact daily evidence groups: original score snippet, generated transcription, local audio, local video, heat-map context, and specific observation.
Current accepted transcription	1 score-linked six-note Staff 4 measure-window is accepted: source Eb5 Eb5 C5 Eb5 Eb5 Eb5, detected audio D#5 D#5 C5 D#5 D#5 D#5, exact MIDI 75 75 72 75 75 75. It is not a solved long phrase. Audio-checked D/A fragments remain useful note evidence, but they are not score-location matches or long-phrase transcription.
Current accepted score match	1 actual-source score crop is accepted for the Staff 4 Eb5 Eb5 C5 Eb5 Eb5 Eb5 window. The 2026-05-03 Bb4 D5 C5 Bb4 D5 Scherzo-Tarantelle crop, the old A4/A5 single-note anchors, and the previous D–B-flat–G–D score extraction are all rejected fixtures. The right-2 Eb5 Eb5 C5 Eb5 Eb5 Eb5 C5 source lane is still not accepted until its full seven-note score/audio/crop/truth state is audited; 951 local score glyph candidates, 214 staff-position notehead hypotheses, 8 staff review packets, and source-verification targets are queued for truth review but are not accepted score evidence.
Current archive coverage	234 h 57 min uploaded practice video across 48 strict-ledger videos and 41 practice days; 2 h 28 min checked; 2 h 21 min measured active violin; full chronological coverage still pending.
Current estimate	223 h 5 min total active practice estimated from checked windows; provisional and sample-biased until the full chronological scan covers the archive.
Current user burden to remove	Alan is still forced to inspect whether a note is real, whether a score box points to the correct pitch, whether audio matches notation, and whether practice time means playing rather than uploaded duration.
First tracker update	This PDF now defines the full implementation plan and becomes the required progress record for future Curtis work.

Progress log

2026-05-13 11:57 AM EDT code update Backend coverage ledger, evidence-correction route, and benchmark-fixture scaffold added. Checked video time now includes sampled non-playing windows; total practice time remains measured active playing only. Live deployed coverage now reports 213 h 23 min uploaded video, 2 h checked video, 53 min 14 s measured active practice, and 211 h 23 min unchecked. Wrong score-note corrections can be stored as rejected benchmark cases. Full archive scan and long-phrase transcription remain unsolved.

2026-05-13 12:33 PM EDT code update Low-cost active-practice scan route implemented. The new scan persists active violin-playing sub-intervals from stored violin-positive media samples, keeps checked non-playing samples separate from practice time, exposes `/api/curtis/active-practice-scan`, adds an active-scan run endpoint, queues remaining strict-ledger windows for owner media capture, and feeds persisted active intervals into daily records and the coverage ledger. Local unit tests verify active sub-interval extraction, negative-sample withholding, and coverage accounting.

2026-05-13 12:59 PM EDT deployed scan update The active-scan endpoint was deployed and run against the live Curtis state. It persisted 214 active sub-intervals from 80 stored media-sample results, representing 41 min 42 s of scan-derived active audio, and queued 250 strict-ledger windows for later owner-media capture. The coverage ledger reports 1 h 57 min 24 s measured active practice from 2 h checked and a 208 h 46 min provisional archive estimate. A regression fix now caps active practice by checked source coverage and makes active-scan results supersede noisy transcription-duration evidence where scan results exist. The full 213 h 23 min archive scan and long-phrase transcription remain unsolved.

2026-05-13 03:26 PM EDT queue update The owner-media sync now consumes the active-scan pending-window queue before choosing expansion or blind sample windows. The active-scan status endpoint can expose a bounded pending-window preview, and the run endpoint returns a compact scan/coverage payload rather than the full operations document. Regression tests now verify that queued active-practice windows are selected first and that pending-window previews preserve total pending counts while limiting returned rows. This does not solve long-phrase transcription; it closes

the next M1 plumbing gap so the chronological archive scan can keep advancing without Alan manually choosing windows.

2026-05-13 03:37 PM EDT live owner-sync update A one-window owner-sync pass uploaded one cached violin-positive 5/3 sample, Njh8_zq9_DM-10725 from *10725-10815. The follow-up active scan selected that new sample and recorded one new active interval. Live coverage is now 81 active-scan sample results, 215 active sub-intervals, 42 min 9 s scan-derived active audio, 2 h 1 min 30 s checked video, 1 h 58 min 54 s measured active practice, and a 208 h 49 min provisional estimate. The first chronological pending queue item remains violin 1 at *0-90; this run advanced cached-media coverage, not the chronological queue.

2026-05-13 04:05 PM EDT completion tracker and queue-only update Curtis now exposes a bottom-page transcription completion tracker backed by a 100-point implementation gate checklist. The tracker reports archive coverage, accepted long phrases, exact score windows, scan queue state, completed gates, remaining gates, and the next concrete action. A queue-only owner-sync mode was added so the active-scan pending-window list can be consumed without falling back to cached positives, expansion windows, or blind sampling. A queue-only run uploaded otHfHMgDo2g-0 from violin 1 at *0-90; the sample was classified as not_violin_or_unclear and therefore did not add practice time. The follow-up active scan reports 82 sample results, 72 active-violin samples, 10 checked non-violin samples, 215 active sub-intervals, 2 h 1 min 30 s checked video, 1 h 58 min 54 s measured active practice, and a 208 h 49 min provisional estimate. The next chronological pending window is violin 1 at *90-180.

2026-05-13 05:44 PM EDT queue resume and practice-time ceiling update The next queue-only owner-sync attempt captured violin 1 windows *90-540. The wrapper timed out after the uploads, revealing that cached pending-window media was excluded from queue-only resume. The owner-sync helper now resumes cached queue windows, including cached non-playing windows, and tests cover that path. The follow-up active scan reports 87 sample results, 77 active-violin samples, 10 checked non-violin samples, 238 active sub-intervals, 2 h 3 min checked video, 1 h 55 min 30 s measured active practice, and a 200 h 23 min provisional estimate. Practice time is now hard-capped by checked source coverage, so unchecked transcription remnants cannot make active practice exceed checked video. The next chronological pending window is violin 1 at *540-630.

2026-05-13 06:01 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded otHfHMgDo2g-540 from violin 1 at *540-630. The sample was classified as violin_positive with a sampler score of 68.6. The follow-up active scan selected that sample, recorded 4 new active sub-intervals, and reports 88 sample results, 78 active-violin samples, 10 checked non-violin samples, 242 active sub-intervals, 2 h 4 min 30 s checked video, 1 h 57 min measured active practice, and a 200 h 32 min provisional estimate. The live accounting check passes because active practice remains below checked video. The next chronological pending window is violin 1 at *630-720.

2026-05-13 06:26 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded otHfHMgDo2g-630 from violin 1 at *630-720. The sample was classified as violin_positive with a sampler score of 60.3. The follow-up active scan selected that sample, recorded 5 new active sub-intervals, and reports 89 sample results, 79 active-violin samples, 10 checked non-violin samples, 247 active sub-intervals, 2 h 6 min checked video, 1 h 58 min 30 s measured active practice, and a 200 h 41 min provisional estimate. The live accounting check passes because active practice remains below checked video. The next chronological pending window is violin 1 at *720-810.

2026-05-13 07:55 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded otHfHMgDo2g-720 from violin 1 at *720-810. The sample was classified as violin_positive with a sampler score of 55.2. The follow-up active scan selected that sample, recorded 3 new active sub-intervals, and reports 90 sample results, 80 active-violin samples, 10 checked non-violin samples, 250 active sub-intervals, 2 h 6 min checked video, 2 h measured active practice, and a 203 h 14 min provisional estimate. The live accounting check passes because active practice remains below checked video. The next chronological pending window is violin 1 at *810-900.

2026-05-13 09:25 PM EDT scan and benchmark update Queue-only owner sync consumed the next two chronological violin 1 windows, *810-900 and *900-990. Both were classified as violin_positive. The follow-up active scan reports 92 sample results, 82 active-violin samples, 10 checked non-violin samples, 257 active sub-intervals, 2 h 10 min checked video, 2 h 3 min measured active practice, and a 201 h 7 min provisional archive estimate. The next chronological pending window is violin 1 at *990-1080. The same pass stored three real user-reviewed score-match

failures as benchmark corrections: A4 displayed against a B-position score crop, A4 displayed against a G4 crop, and A5 displayed against a G5 crop. These benchmark fixtures moved the implementation tracker from 29 percent to 31 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-13 10:05 PM EDT scan and exact-percent update Queue-only owner sync consumed violin 1 windows *990-1080 and *1080-1170. Both were classified as `violin_positive`. The follow-up active scan reports 94 sample results, 84 active-violin samples, 10 checked non-violin samples, 269 active sub-intervals, 2 h 13 min checked video, 2 h 6 min measured active practice, and a 201 h 24 min provisional archive estimate. The next chronological pending window is violin 1 at *1170-1260. The rounded implementation label remains 31 percent, but exact weighted progress moved to 31.13 percent and is now displayed so small coverage gains are visible. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:14 PM EDT scan and precision update Queue-only owner sync consumed violin 1 windows *1170-1260 and *1260-1350. The first was classified as `violin_positive`; the second was classified as `not_violin_or_unclear`, which adds checked non-playing coverage without adding practice time. The follow-up active scan reports 96 sample results, 85 active-violin samples, 11 checked non-violin samples, 283 active sub-intervals, 2 h 16 min checked video, 2 h 7 min measured active practice, and a 199 h 19 min provisional archive estimate. The next chronological pending window is violin 1 at *1350-1440. The implementation tracker now computes exact weighted progress from unrounded gate points, so the live label should read 31.128 percent after deployment. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:38 PM EDT scan update Queue-only owner sync consumed violin 1 windows *1350-1440 and *1440-1530. Both were classified as `violin_positive`. The follow-up active scan selected both samples, recorded 11 new active sub-intervals, and reports 98 sample results, 87 active-violin samples, 11 checked non-violin samples, 294 active sub-intervals, 2 h 19 min checked video, 2 h 10 min measured active practice, and a 199 h 37 min provisional archive estimate. The next chronological pending window is violin 1 at *1530-1620. Exact weighted implementation progress is now 31.131 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:55 PM EDT scan update Queue-only owner sync consumed violin 1 windows *1530-1620 and *1620-1710. The first was classified as `not_violin_or_unclear`; the second was classified as `violin_positive`. The follow-up active scan selected both samples, recorded 6 new active sub-intervals, and reports 100 sample results, 88 active-violin samples, 12 checked non-violin samples, 300 active sub-intervals, 2 h 22 min checked video, 2 h 12 min measured active practice, and a 197 h 40 min provisional archive estimate. The next chronological pending window is violin 1 at *1710-1800. Exact weighted implementation progress is now 31.134 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 12:09 AM EDT scan update Queue-only owner sync consumed violin 1 windows *1710-1800 and *1800-1890. Both were classified as `violin_positive`. The follow-up active scan selected both samples, recorded 13 new active sub-intervals, and reports 102 sample results, 90 active-violin samples, 12 checked non-violin samples, 313 active sub-intervals, 2 h 25 min checked video, 2 h 15 min measured active practice, and a 197 h 59 min provisional archive estimate. The next chronological pending window is violin 1 at *1890-1980. Exact weighted implementation progress is now 31.136 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 12:35 AM EDT scan update Queue-only owner sync consumed violin 1 windows *1890-1980 and *1980-2070. Both were classified as `violin_positive`. The follow-up active scan selected both samples, recorded 9 new active sub-intervals, and reports 104 sample results, 92 active-violin samples, 12 checked non-violin samples, 322 active sub-intervals, 2 h 28 min checked video, 2 h 18 min measured active practice, and a 198 h 18 min provisional archive estimate. The next chronological pending window is violin 1 at *2070-2160. Exact weighted implementation progress is now 31.139 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 03:00 AM EDT scan update Queue-only owner sync consumed violin 1 windows *2070-2160 and *2160-2250. Both were classified as `violin_positive`. The follow-up active scan folded both samples into the checked ledger and reports 106 sample results, 94 active-violin samples, 12 checked non-violin samples, 331 active sub-intervals, 2 h 31 min checked video, 2 h 21 min measured active practice, and a 198 h 36 min provisional archive estimate. The next chronological pending window is violin 1 at *2250-2340. Exact weighted implementation progress is now 31.142 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 03:23 AM EDT one-measure accelerator update Curtis no longer hardcodes accepted long phrases to zero. The tracker now derives accepted long-phrase count from daily match groups that pass all of the following gates: score location verified, exact score-location status, at least a five-note phrase run, local playable media, and not merely a single-pitch anchor. Regression tests now prove that a verified one-measure symbolic-score phrase with media increments the long-phrase count, while single-note anchors, unverified reference matches, and matches without media do not. This is implementation progress toward a faster path, not solved musical evidence yet: the live percent remains 31.142 percent until Curtis has a verified symbolic score measure or more scanned archive coverage. The fastest route to visible musical progress is now to verify one Scherzo-Tarantelle symbolic score measure and run the existing phrase matcher across the hidden detected note series already stored for 2026-05-03.

2026-05-14 04:32 AM EDT source-score acceleration update Curtis now carries the Wieniawski Scherzo-Tarantelle solo score PDF as a local source asset rather than depending on weak scanned crops or a network fetch during matching. The score-asset layer reports whether a local source PDF is present, the daily score-reference audit counts local source PDFs separately from symbolic notes, and the 100-point tracker awards source-truth progress for a local source score without counting it as an accepted phrase. Regression tests verify that this raises the score-source readiness state while keeping accepted long phrases, exact score windows, and symbolic score notes at 0. This is the necessary acceleration layer for long-phrase work: the next step is no longer finding a score image, but converting one local source-score measure into verified symbolic notes and running the existing phrase matcher over stored detected note series.

2026-05-14 05:23 AM EDT live symbolic-measure update The deployed Curtis API reported 43.642 percent exact progress, 1 provisional measure match, 1 provisional score-aligned window, 0 accepted long phrases, 1 local score source, and 8 symbolic score notes counted across the two Wieniawski practice-day targets. This state is now superseded: the later source-truth correction rejected the provisional D A# G D source-score claim because the local IMSLP solo part did not support that score-note sequence. The useful implementation gain from this pass was the source-symbolic note gate and score/audio/video/transcription lane, not the specific provisional score phrase.

2026-05-14 05:40 AM EDT source-target accounting update The deployed Curtis API reported 45.142 percent exact progress after the tracker began counting configured source score targets from the reference-training state instead of only local PDF files. That accounting layer remains useful, but the musical evidence count from this pre-correction state is superseded by the 07:52 AM source-truth correction. This makes the progress meter more honest and removes a hidden undercount, but it does not solve phrase transcription; the next implementation step is extending corrected accepted symbolic measure gates into longer matched note runs.

2026-05-14 06:52 AM EDT phrase-candidate ranking update The deployed Curtis API reported 47.642 percent exact progress after the matcher stopped returning the first scan-order matches and instead ranked all detected-series matches by exact score status, score-derived status, distinct pitch-class content, and run length. That state exposed 12 pitch-sequence groups, 1 phrase candidate, 1 provisional measure match, 1 provisional score-aligned window, and 0 accepted long phrases. The highest-ranked pending phrase candidate was D D D# D D A# G. This candidate was not accepted score evidence; it was a focused source-score verification target. The later strict evidence gate supersedes this provisional count.

2026-05-14 07:08 AM EDT tracker visibility update Curtis now returns the highest-ranked pending phrase target as structured tracker state and displays it in the implementation card detail as **pending**: D D D# D D A# G. The visible interface therefore shows not only that one phrase candidate exists, but also which detected pitch-class sequence should be source-score verified next. This does not add accepted musical evidence or change the 47.642 percent full-solve score; it removes a product friction point by making the next verification target visible without presenting it as a solved score match.

2026-05-14 07:52 AM EDT source-truth correction update The Wieniawski symbolic source opening was corrected against the local IMSLP solo-violin score. The old accepted D A# G D match was demoted because the source phrase is D C Bb D C Bb; it can remain only as an unverified reference-sequence candidate, not accepted score evidence. The deployed provisional groups matched detected pitch classes A# D C A# to source score pitch classes A# D C A#, displayed a source-score crop for that phrase, and retained generated notation only as fallback. The tracker then reported 48.642 percent, 2 provisional measure matches, 2 provisional score-aligned windows, 12 pitch-sequence groups, 1 pending phrase candidate, 3 score targets, 1 local score PDF, 6 unique symbolic score notes,

1 source snippet candidate, and 0 accepted long phrases. The later strict evidence gate supersedes these provisional score counts.

2026-05-14 08:29 AM EDT first source-backed phrase update The Scherzo-Tarantelle source excerpt was extended by one verified score note, from D C Bb D C Bb to D C Bb D C Bb D, using a widened crop from the local IMSLP solo-violin PDF rather than the annotated teaching image. That seven-note source phrase let the stored 2026-05-03 detected series match the contiguous five-note run A# D C A# D against source score notes Bb4 D5 C5 Bb4 D5. The deployed API then reported 55.642 percent exact progress, 1 provisional long-phrase candidate, 2 provisional measure matches, 2 provisional score-aligned windows, 12 pitch-sequence groups, 7 unique symbolic score notes, and 1 source snippet candidate. The useful gain was the source-backed path; the later visual review rejected the crop as accepted score evidence, so the strict current state is again 0 accepted long phrases.

2026-05-14 01:50 PM EDT score-coordinate heat-map update Provisional symbolic score phrase matches were connected to score-coordinate heat-map fragments. The 2026-05-03 Scherzo-Tarantelle phrase A# D C A# D was mapped to mm. 2-4 with source score notes Bb4 D5 C5 Bb4 D5, local video, local audio, and an IMSLP score crop. The deployed API reported 58.136 percent exact progress, 1 provisional long-phrase candidate, 2 provisional measure matches, 2 provisional score-aligned windows, 2 provisional score-coordinate heat-map fragments, 12 pitch-sequence groups, 7 unique symbolic score notes, and 1 source snippet candidate. This was a heat-map and evidence-organization advance; the later strict evidence gate blocks those fragments from counting as accepted evidence.

2026-05-14 02:15 PM EDT source-verification target update Curtis now separates the next longer note-run target from accepted score evidence. The tracker exposes one source-verification target, D D D# D D A# G, only because it is a seven-note run from local playable practice media with enough pitch diversity to be worth checking against the score. It remains reference-audio evidence, not accepted score evidence. The source target appears in the tracker and API so the next pass can verify exact local IMSLP score notes and coordinates instead of slowly searching from scratch or accidentally promoting a wrong crop. The deployed tracker state then included provisional score artifacts, but the current strict evidence gate counts 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, and 0 verified source snippets until source review passes.

2026-05-14 02:51 PM EDT source-target triage update Curtis now checks four-note-or-longer source targets against the verified symbolic score map before promotion. The current top target D C G A was compared against the verified source excerpt D C A# D C A# D. The best contiguous overlap is two notes, so the target is recorded as `source_score_sequence_not_found` and remains reference-audio evidence only. The tracker then reported 58.633 percent exact progress, 3 source-target score checks, 0 verified source targets, and the next action: extend the verified IMSLP symbolic score map before promoting this sequence. This improves the long-phrase pipeline by making measure-sized candidates fail fast and explicitly instead of appearing as misleading score matches.

2026-05-14 03:49 PM EDT score-map workbench update Curtis now has an explicit source-map acceleration layer for the local IMSLP Wieniawski solo part. A deterministic extraction tool renders page 2 of the local score PDF, detects staff-line groups, removes staff lines, extracts connected score glyph candidates, and writes a JSON verification queue. The current queue contains 779 unaccepted score glyph candidates across 7 staves. These candidates are not pitch labels, not accepted score notes, and not displayable score evidence. They exist to make the next work cycle faster: verify candidate glyphs against the local PDF, convert only verified notes into MusicXML, then run source-target matching against verified symbolic notes. The tracker now reports 59.633 percent exact progress after awarding score-truth credit for the verification queue while preserving 0 verified source targets for the current D C G A source target.

2026-05-14 04:24 PM EDT note-hypothesis acceleration update The source-map workbench now filters the local IMSLP glyph queue into likely notehead candidates and assigns staff-position pitch hypotheses using the G-minor key signature. The current queue contains 172 unaccepted score-note hypotheses across 7 staves in addition to the 779 raw glyph candidates. These hypotheses are explicitly not accepted score evidence: they use staff position only, may include false positives, and cannot create a score snippet, pitch label, or match card until source-reviewed into MusicXML. The implementation value is speed. The next pass can review a note-labeled queue instead of starting from anonymous glyph boxes, which should make extending the Scherzo-Tarantelle symbolic map to full measures and longer phrases much faster. The tracker now reports 61.133 percent exact progress after awarding limited score-truth credit for the note-hypothesis queue while preserving 0 verified source targets for the current D C G A source target.

2026-05-14 05:31 PM EDT staff-review packet update The score-map detector was changed from a single-threshold staff finder to a multi-threshold merged staff finder after review showed that the previous queue skipped a source staff. Page-2 coverage now reports 951 unaccepted score glyph candidates and 214 unaccepted notehead hypotheses across 8 staves. The same pass writes 8 staff-level review-packet images under the local score assets folder, with each packet showing the scanned IMSLP staff plus numbered hypothesis boxes. The packets are not accepted score evidence and do not change any match card by themselves; they exist to make the next MusicXML pass faster and less error-prone by turning source review into a bounded staff-by-staff checklist. The deployed tracker now reports 64.135 percent exact progress after awarding score-truth credit for complete source-review packet generation while preserving 0 verified source targets for the current D D D# D D A# G source target, whose verified symbolic-score overlap is still only 1/7.

2026-05-14 06:15 PM EDT score/transcription visual-agreement update The public 2026-05-03 match had a real failure: the score panel showed a broad opening source crop even though the claimed transcription phrase was the narrower five-note slice Bb4 D5 C5 Bb4 D5. Curtis then added a separate score-display gate: a score panel could render an actual source crop only when the crop's reference start and end exactly matched the phrase, and broad covering crops were withheld. The symbolic matcher also began requiring MIDI-level equality between detected notes and parsed score notes, not only pitch-class similarity, before it could create a score-visual lock. The deployed tracker reported 64.878 percent exact progress and provisional score locks, but later visual review showed the crop still was not trustworthy enough; the current strict gate counts this phrase as pending, not accepted.

2026-05-14 07:04 PM EDT actual-source score-gate update The previous visual-agreement fix still left one unacceptable loophole: a generated notation SVG could appear in the accepted score panel even though Alan asked for the actual piece. Curtis therefore required a score panel to use a non-generated source-score crop for the same reference range. The five-note 2026-05-03 Scherzo-Tarantelle match gained an exact-range source crop for Bb4 D5 C5 Bb4 D5; generated notation remained only as the transcription rendering. That deployment expected provisional actual-source score progress, but the later musician review rejected the crop and the current strict evidence gate blocks it from creating an accepted measure, long-phrase count, exact score-aligned window, score-coordinate heat-map fragment, actual-source score lock, or verified source snippet.

2026-05-14 09:45 PM EDT score-spelling visual-context update The latest reported mismatch exposed another gate that had been too weak: a transcription could be score-spelled as Bb4 D5 C5 Bb4 D5 in data while the rendered notation panel omitted the source key signature, making the first and fourth notes look like unqualified natural-position B notes to a reader. Curtis now carries the source key signature into score-matched transcription rendering and requires both `scoreVisualRangeAgreement` and `scoreSpellingAgreement` before an actual-source score crop can count as accepted score evidence. The Scherzo-Tarantelle match remains a five-note fragment, but the visible transcription and visible score must now agree in musical spelling context rather than only MIDI or pitch class. This is a stricter acceptance gate, not a new long-phrase solve.

2026-05-15 12:29 PM EDT exact-note truth-gate update The reported failure mode was not only visual range: a score match could still be treated as acceptable if the pitch class agreed while the written note/register did not. The truth workbench now stores MIDI sequences for detected, accepted, and score notes; accepted score-truth items reject audio-score mismatches at exact note-and-octave level; and score-truth items cannot be accepted with score notes that omit octave/register. Single-note score anchors also require an explicit exact-note verification flag before the UI can render them. Regression tests now cover accepted A4–D5 truth, A4-versus-A5 rejection, score truth without octave rejection, exact pitch-anchor display, and visual-only pitch-anchor rejection. This does not create a new accepted phrase; it prevents Curtis from ever calling a pitch-class-only or visually guessed crop solved evidence.

2026-05-14 10:52 PM EDT visual-crop demotion update Alan's follow-up review showed that the notes visible in the 2026-05-03 source crop still did not match the transcription. Curtis now rejects the Scherzo-Tarantelle source crops as review material unless the visible noteheads, register, written spelling, and exact range are independently verified against the transcription. The five-note 5/3 phrase remains a symbolic phrase candidate, but it no longer creates an accepted score panel, accepted measure match, accepted long phrase, exact score-aligned window, score-coordinate heat-map fragment, score-visual lock, actual-source score lock, or verified source snippet. This is a regression lock: wrong score evidence is removed rather than cosmetically relabeled.

2026-05-15 12:45 AM EDT truth-workbench update Curtis now has a persistent truth-workbench API and compact implementation-status surface for accepted, pending, and rejected audio-score-transcription evidence. A truth item can store the source clip, practice day, piece, detected notes,

accepted notes, score notes, score source, score location, score image, and gate state. Accepted truth items are rejected at write time if the accepted audio note sequence and score note sequence disagree. The backend and frontend score gates also now require explicit visible-note-sequence verification before a score panel can count as a score-visual lock, actual-source score lock, accepted measure match, or accepted long phrase. This prevents the exact failure Alan found: a visually wrong score crop cannot count just because the metadata says the pitch sequence should match. It still has 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, 0 score-visual locks, 0 actual-source score locks, and 0 accepted truth clips.

2026-05-15 03:20 PM EDT exact-source phrase attempt Curtis temporarily promoted one actual-source Scherzo-Tarantelle phrase crop, Bb4 D5 C5 Bb4 D5 from the local IMSLP page-2 solo-violin scan, aligned to the symbolic score range mm. 2–4. The attempt boxed five matched noteheads and stored source page pixel coordinates and exact visible note-plus-octave sequence. This pass was useful because it proved the crop extraction path could render actual score material, but it was later rejected by visual review because the score notes and boxes were not trustworthy enough to count as accepted evidence.

2026-05-15 03:40 PM EDT superseded deployment verification The deployed Curtis site served cache version 20260515-exact-source-phrase on deployment 4fb78ce1. That deployment still showed one score group for 2026-05-03, with detected notes A#4 D5 C5 A#4 D5, score-spelled transcription Bb4 D5 C5 Bb4 D5, source score notes Bb4 D5 C5 Bb4 D5, the exact-source crop asset, local video sample Njh8_zq9_DM-9465, and local audio from 6.618–7.674 s inside that sample. This state is now superseded by the 05:00 PM EDT strict evidence-gate pass because the displayed score crop did not survive musician visual review. The live practice-time surface reported 234 h 57 min uploaded archive video, 2 h 28 min checked video, 2 h 21 min measured active violin, 223 h 5 min estimated total active practice, and 232 h 28 min unchecked video. The estimate remains provisional until the full chronological active-practice scan covers every archive window.

2026-05-15 05:00 PM EDT strict evidence-gate update Curtis now rejects the 2026-05-03 Scherzo-Tarantelle source crop as accepted evidence unless the full truth chain passes: source note boxes centered on the visible noteheads, visible score notes matching the claimed written notes and octaves, transcription notes matching the paired audio, transcription notes matching the score notes, and an

accepted truth-workbench item tying the audio, transcription, and score together. Backend, frontend, tracker, and tests now require those fields before a score snippet, accepted measure match, exact score-aligned window, actual-source score lock, or long-phrase count can appear. The deployed tracker reports 43.126 percent exact weighted progress, 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 actual-source score locks, and 0 verified source snippets until a crop survives this stricter review.

2026-05-15 04:01 PM EDT practice-time API unification The daily-record API now consumes the same active-practice coverage ledger used by `/api/curtis/active-practice-coverage` before publishing top-level and per-day practice totals. This removes the stale split where one public route could show older transcription-derived totals while the coverage ledger showed 2 h 28 min checked video, 2 h 21 min measured active violin, 223 h 5 min estimated total active practice, and 232 h 28 min unchecked video. A regression test now verifies that daily records inherit canonical checked, active, estimate, unmeasured, and per-day scan totals from the coverage ledger.

2026-05-15 06:34 PM EDT strict-source correction attempt Curtis temporarily promoted a corrected actual-source crop for the same 2026-05-03 Scherzo-Tarantelle phrase. The detected sequence was A#4 D5 C5 A#4 D5; the displayed spelling was Bb4 D5 C5 Bb4 D5; the score-side sequence was the same octave-aware note sequence; and the score panel used a non-generated crop from the local Scherzo-Tarantelle solo part. This attempt is now superseded by Alan's later review: the phrase is rejected and cannot count as an accepted measure-level match, exact score-aligned window, score-coordinate heat-map fragment, score-visual lock, actual-source score lock, verified source snippet, or accepted five-note phrase.

2026-05-15 08:37 PM EDT fast-transition candidate update The next blocker was not another UI issue: the broader pYIN/onset stream in the 2026-05-03 Scherzo-Tarantelle window still collapses large parts of fast playing into repeated D/B-flat/G material, so a longer source-score phrase cannot be honestly promoted. Curtis now adds a hidden YIN frame-transition trace to every transcription run. This trace follows fast frame-level pitch changes and stores them as `scoreMatchCandidateNotes`; it is never rendered as accepted notation by itself. The daily matcher can search those hidden candidates against parsed source-score notes, but accepted score evidence still requires exact MIDI equality, exact written score spelling, visible source-score crop agreement, local audio/video, and no uncertain

octave-corrected detected notes. Regression tests now prove that hidden transition candidates can feed the strict score matcher while uncertain octave-corrected candidates are rejected. A local check on the 5/3 Njh8_zq9_DM-9465 clip produced 174 hidden fast-transition candidate notes with 8 pitch classes. After the five-note rejection, those candidates remain search data only and do not create accepted evidence. This is implementation progress toward long phrases, not a solved long phrase.

2026-05-15 08:48 PM EDT notation-renderer correction The web notation renderer now moves the treble-clef baseline upward and re-anchors flat key-signature glyphs so B-flat and E-flat render on the intended staff positions. This fixes a display-layer source of visible wrongness, but it does not count as transcription progress unless the paired audio and source-score sequence pass the strict acceptance gate.

2026-05-15 09:07 PM EDT targeted-transcription update Live v11 reprocessing exposed a workflow bottleneck: the public transcribe endpoint could only walk the stale media queue in fixed eight-sample passes, so a known evidence clip could remain stale while unrelated samples were regenerated first. The endpoint now accepts a bounded `sampleId` query parameter and an optional bounded `limit`, allowing a specific local media sample to be regenerated under the current pipeline without bypassing violin-positive checks, hidden-candidate gating, uncertainty rejection, or source-score acceptance rules. This is required for faster long-pharse iteration because failures can be retested directly instead of waiting through broad queue churn.

2026-05-15 09:20 PM EDT live targeted-reprocess verification The deployed endpoint was run against Njh8_zq9_DM-9465. The live tracker then reported 67.375 percent exact weighted progress, 1380 hidden transition-trace candidates, 1 accepted long-pharse counter item, 1 accepted measure-match counter item, and 1 exact-source visible match group on 2026-05-03. Browser verification of the public site showed one visible group: Bb D C Bb D in the score panel, local video/audio from 2:37:51–2:37:52, and transcription Bb4 D5 C5 Bb4 D5 / 1.1s. Alan's later review rejected that group, so the retained progress is the targeted endpoint and hidden-candidate instrumentation, not the accepted phrase.

2026-05-15 09:24 PM EDT post-deploy state check A post-deploy API read showed that live background processing had advanced the implementation tracker to 68.125 percent exact weighted

progress and 2476 hidden transition-trace candidates. That one-group state is now superseded by the 10:05 PM EDT rejection lock. The extra hidden candidates improve search coverage, but they do not count as solved transcription until a longer exact source-score run is verified and reviewed.

2026-05-15 10:05 PM EDT five-note rejection lock Alan's latest review rejected the five-note Scherzo-Tarantelle phrase itself. Curtis now demotes A#4 D5 C5 A#4 D5 / Bb4 D5 C5 Bb4 D5 from accepted evidence back to candidate-only material. Regression tests now require this exact lane to have zero verified score location, zero verified source snippet, no accepted measure, no accepted long phrase, no score-coordinate heat-map fragment, and no visible accepted score panel. The only items Alan had called usable remain short audio-only D/A detections; the A4/A5 score anchors, D-B-flat-G-D score sequence, and five-note Scherzo phrase are rejected until new independent review proves otherwise.

2026-05-15 10:54 PM EDT second-pass audio-gate update The repeated loop was that fast transition notes and loose reference-audio overlaps could still look like progress before the notes were independently confirmed by audio. A local audit of the rejected Njh8_zq9_DM-9465 clip read the candidate closer to Bb4 D5 Bb4 D5; the displayed middle C5 did not survive the second pass. Curtis now preserves audioAgreement, agreementSources, agreementSourceCount, and detector provenance in daily detected-note series. Hidden scoreMatchCandidateNotes are filtered before score search unless each note has audio agreement, at least one independent agreement source, spectral/onset participation, confidence at or above 0.70, and enough duration to avoid transition glitches. Symbolic score matches and reference-sequence matches now require the same per-note audio gate, and reference-audio pitch-trace matches need at least four notes and three distinct pitch classes. Regression tests now lock the rejected Scherzo phrase when the C5 fails audio agreement, reject unagreed fast candidate notes, reject uncertain octave-corrected candidates, and stop weak two-note reference-audio coincidences from counting as phrase progress. This does not solve long-phrase transcription; it removes a false-positive lane that was making Curtis go in circles.

2026-05-15 11:17 PM EDT candidate-isolation update Curtis now separates source-verification candidates from displayable score matches in the daily-record API. Pitch-sequence candidates can still feed the truth workbench and source-verification target queue, but they no longer populate the visible matchGroups bucket, no longer mark a day as reference-linked, and no longer create heat-map

fragments unless the exact accepted score-evidence gate passes. A new regression test proves that a locally playable Wieniawski source-sequence candidate can be queued with `candidateMatchGroups` while `matchGroups`, accepted score links, and heat-map fragments remain empty. This prevents another circular handoff where Alan has to inspect a displayed “match” and discover that it was only a candidate.

2026-05-15 11:32 PM EDT live candidate-isolation verification The deployed API reported 46.125 percent exact weighted implementation progress, 0 accepted long phrases, 0 accepted measures, 0 exact score-aligned windows, 0 actual-source score snippets, and 0 score-coordinate heat-map fragments. The 2026-05-03 daily record reported `source_verification_pending`, 0 visible `matchGroups`, 6 queued `candidateMatchGroups`, 0 score links, and 0 reference links. The top queued source target is A D D F# D; it remains rejected as accepted score evidence because it does not have a contiguous exact-MIDI run in the corrected source MusicXML.

2026-05-16 source-position and exact-MIDI update The verified symbolic score map itself had a staff-position error. Rechecking the local IMSLP solo-violin page at high resolution and cross-checking the Wieniawski Society bars 5–9 example showed that the opening violin entrance is A5 G5 F5 A5 G5 F5 A5 G#5 F5, not the prior D6 C6 Bb5 D6 C6 Bb5 D6 C#6 B5 map. The MusicXML now stores those nine actual source notes from measures 5–7. The matcher still uses pitch-class overlap for candidate search, but accepted symbolic score matches and source-verification targets now require exact detected MIDI equality, per-note audio agreement, and the same actual-source crop and truth-evidence gates. Regression tests now cover the corrected opening map, a rejected partial exact-MIDI target whose best overlap is only A5 G5, and a positive audio-agreed exact-MIDI phrase G5 F5 A5 G#5 F5 with MIDI 79 77 81 80 77. This still does not accept a live long phrase; it removes the path that let a wrong-staff-position source map or pitch-letter-only target look solved.

2026-05-16 exact-MIDI truth-manifest update The exact-MIDI rule is now a standalone implementation asset instead of only an inline scanner condition. The new `curtis-long-phrase-gold.json` manifest verifies one source-only Wieniawski positive phrase, G5 F5 A5 G#5 F5, and blocks three known false phrase families: the old D6 C6 Bb5 source map, the rejected Bb4 D5 C5 Bb4 D5 lane, and the current queued G5 G5 A#5 A#5 A#5 A#5 A5 target. The shared gate now reports contiguous exact-MIDI overlap, ordered overlap, exact source sequence, and audio-agreement state. The site

exposes the source-positive and blocked-regression counts in the compact solve-status panel. This is not a solved live phrase; it is the regression harness needed so the next phrase attempt cannot reuse a known wrong score map or a pitch-letter-only match.

2026-05-17 Staff 4 source-lane extension The next source-verification step extends the current accepted Staff 4 lane outward instead of starting a new random score match. The verified MusicXML source now carries the adjacent Eb5 C5 notes to the right of the accepted Eb5 Eb5 C5 Eb5 Eb5 window, creating a seven-note source lane Eb5 Eb5 C5 Eb5 Eb5 Eb5 C5. The first implementation state kept that extension source-only because the available paired audio did not yet pass the adjacent-note gate. The later right-1 promotion supersedes that blocker for the sixth note only: Eb5 Eb5 C5 Eb5 Eb5 Eb5 is now accepted source/audio truth, while the right-2 seven-note lane still needs its own audit and truth acceptance. Regression tests require this separation so adjacent source growth cannot silently turn a partial prefix into a longer displayed phrase match.

2026-05-17 Staff 4 phrase-audit packet update Curtis now has a dedicated generator for Staff 4 expansion decisions instead of retrying random score matches. The packet is keyed to the current source/audio lane, extracts the bounded local audio clip, attempts a matching local video clip, preserves stored detected notes, reruns pYIN, YIN, and spectral-peak pitch checks over the same audio, records onset times, writes a pitch-trace SVG, writes a spectrogram SVG, and stores the packet JSON under the runtime audit route. The first blocker packet preserved the Eb5 versus D5 failure. The later right-1 packet can accept the six-note lane only because exact MIDI, source crop, paired audio, and truth state all agree. The next packet must audit the right-2 seven-note lane independently.

2026-05-17 Staff 4 audit-rejection and next-window update The original right-2 Staff 4 audit result is a rejected regression case in the long-phrase truth manifest. The rejected case binds the source target Eb5 Eb5 C5 Eb5 Eb5 Eb5 C5 to the observed paired audio run D#5 D#5 C5 D#5 D#5 D5 D#5; because the sixth note is D5 instead of source Eb5, that exact expansion cannot reappear as accepted evidence. The expansion harness then tested the smaller right-1 window Eb5 Eb5 C5 Eb5 Eb5 Eb5; a later exact audio candidate for that smaller window now passes and is accepted. The rejected right-2 case remains blocked, and the next right-2 attempt must be a fresh audit rather than a reuse of the old failed run.

2026-05-17 Staff 4 adjacent-mining update Curtis now runs a dedicated mining pass over stored May 3 detected audio-note windows for the exact Staff 4 right-1 MIDI sequence 75 75 72 75 75 75 and the longer right-2 sequence 75 75 72 75 75 75 72. This is separate from score display and cannot accept evidence by itself. It reports exact audio-agreed candidates when they exist, otherwise it reports the nearest stored miss and its exact-prefix length. Regression tests cover both outcomes: the current bad run remains `not_found` because it has D5 where the source requires Eb5, while a synthetic raw detected run with D#5 D#5 C5 D#5 D#5 D#5 C5 becomes an exact mining candidate ready for audit.

2026-05-17 03:23 AM EDT Staff 4 source-audio rescanner update Curtis now searches beyond the note windows already stored in the daily record. Starting from the accepted Staff 4 source/audio anchor, the backend locates the original local media sample, extracts a padded source-audio window around the accepted clip, reruns the note detector on that fresh audio, shifts the detected event times back into the source-video timeline, and adds those runs to both the anchored-expansion harness and the adjacent-mining pass. The cache key includes the media path, file size, modification time, source window, and transcription-pipeline version, so stale detector output is not silently reused after source media or extractor changes. This still cannot promote a longer phrase by itself: a rescanned run can only become a review candidate when its exact MIDI sequence matches the verified Staff 4 source lane, and it still needs paired-audio review, source crop agreement, local video/audio, and truth acceptance before it displays as a score match. The regression suite now proves that an exact 75 75 72 75 75 75 72 phrase found only in a fresh source-audio rescan enters `ready_for_truth_review` and the mining pool, while failed or missing source media remains blocked instead of becoming visible evidence.

2026-05-17 Staff 4 anchor-persistence update The accepted Staff 4 lane no longer depends on the current daily display retaining an accepted `matchGroups` card. Curtis now derives the anchor from the long-phrase truth manifest when the daily record has no visible accepted match group: source id `wieniawski-staff4-eb-eb-c-eb-eb-symbolic-v1`, source MIDI 75 75 72 75 75, practice day 2026-05-03, sample `Njh8_zq9_DM-8835`, source window `*8835-8925`, and local phrase timing. The source-audio rescanner, expansion harness, and adjacent-mining pass all call the shared anchor resolver and deduplicate the manifest anchor against an existing visible accepted anchor. A regression test now reproduces the live failure directly: the May 3 record has zero visible `matchGroups`, yet

1077 source-audio rescan still runs, phrase expansion still has an anchor, adjacent mining still searches
1078 the exact Staff 4 MIDI lane, and `no_staff4_anchor` does not return. This protects the fixed-lane
1079 workflow; it does not promote any new phrase.

1080 **2026-05-17 12:56 PM EDT Staff 4 tiled-rescan update** Curtis now widens that source-audio
1081 search from one padded clip to a bounded tiled plan over the same source sample. For the accepted
1082 *8835–8925 Staff 4 anchor, the backend generates the anchor-core tile plus four overlapping rightward
1083 tiles, each no longer than 18 seconds. Each tile has its own cache key, audio extract, detector run,
1084 shifted source-window timing, and merged audio-note run. Regression tests now require five Staff 4
1085 rescan windows and prove that exact MIDI found only in those tiled source-audio runs still enters the
1086 expansion/mining pool without becoming visible accepted evidence.

1087 **2026-05-17 01:09 PM EDT Staff 4 anchor-reproduction update** Curtis now checks whether the
1088 fresh current-detector source-audio rescan can reproduce the accepted Staff 4 anchor before treating the
1089 widened scan as usable phrase-extension evidence. The live tiled pass produced five source-window
1090 tiles, four note runs, and 286 primary-plus-candidate note events, but adjacent mining still found
1091 zero exact Staff 4 right-extension candidates after 1547 stored/rescanned windows. The same pass
1092 also did not reproduce the accepted five-note anchor 75 75 72 75 75. The tracker now records
1093 `not_reproduced` instead of implying that anchor persistence equals fresh detector agreement. This
1094 changes the next implementation step: calibrate segmentation or accept a new current-detector anchor
1095 before expanding the Staff 4 phrase.

1096 **2026-05-16 09:07 PM EDT gold-review-loop update** Curtis now has a real review loop instead
1097 of only hidden candidates and scattered correction endpoints. The backend builds a queue from
1098 detected audio-note series and candidate score groups, choosing reviewable note windows that prefer
1099 audio-agreed, spectrally supported, non-repeated phrase material over repeated-note sludge. The
1100 new `/api/curtis/gold-review` route exposes accepted, rejected, pending, queue, score-ready,
1101 and evidence-ready counts. The `/api/curtis/gold-review/items` write path accepts pending,
1102 accepted, or rejected labels and mirrors accepted/rejected labels into the truth workbench. Audio-only
1103 accepted labels become gold audio truth; accepted score labels are refused unless the accepted
1104 audio notes and score notes match exactly by note name, octave, and MIDI sequence and include a
1105 source score coordinate. The site now includes a compact review surface with local video, audio,

detected notation, editable accepted notes, optional score notes/location, and accept/reject actions. 1106
 The solve-status tracker includes gold-label and queue counts. This still leaves live long phrases at 1107
 zero; it creates the review data path required to make future phrase work accumulate instead of cycling 1108
 through unverifiable candidates. 1109

2026-05-16 09:36 PM EDT playable-window guard update Live browser verification found a real 1110
 media failure in the new review loop: one queued gold item pointed at a local audio extraction window 1111
 longer than the playback endpoint permits. Curtis now rejects or shortens review-note windows before 1112
 they enter the queue so every published gold candidate has a playable local clip route. A regression 1113
 test covers a long-spaced detected series and proves the queued candidate stays within the endpoint's 1114
 playback limit. This is not musical transcription progress by itself; it is an evidence-loop guard so the 1115
 review queue cannot ask Alan to label a broken clip. 1116

5.2 Definition of solved 1117

Long-phrase transcription is solved for Curtis only when the system can take Alan's long-form practice 1118
 archive and produce evidence groups that a musician can trust without manually checking every note. 1119
 The complete goal is not merely a model that prints plausible notation. The solved system must satisfy 1120
 all of the following conditions. 1121

1. **Full archive accounting.** Every practice video from the strict practice ledger is assigned to 1122
 the correct practice day, has exact uploaded duration, and has a measured or explicitly pending 1123
 active-practice state. 1124
2. **Active violin time.** Total practice time means intervals where Alan is actually playing violin. 1125
 Talking, tuning, setup, walking, camera adjustment, silence, scrolling, and other non-playing 1126
 intervals are excluded. 1127
3. **Local evidence playback.** Every accepted transcription snippet has a playable local video clip 1128
 and audio clip in the Curtis site. Accepted evidence does not rely on a YouTube embed. 1129
4. **Accurate note events.** The transcribed notes match the paired audio, including repeated notes, 1130
 fast scalar runs, arpeggios, shifts, string crossings, rests, and obvious restarts. A fast passage 1131
 must not collapse into a false repeated pitch stream such as D D D D. 1132

- 1133 5. **Rhythm and timing.** The system estimates onsets, durations, rests, tempo changes, slow-practice
1134 stretches, restarts, and repeated attempts. When rhythm is uncertain, the notation must mark
1135 uncertainty rather than invent a clean answer.
- 1136 6. **Professional notation.** Displayed notation uses a real notation engine or a notation-quality
1137 rendering path. Treble clef placement, staff scaling, note placement, stems, beams, rests, spacing,
1138 key signature, and measure layout must look like real sheet music, not a hand-drawn mockup.
- 1139 7. **Score-source truth.** A score snippet can appear only when the score-side notes are source-
1140 confirmed. Scanned images without a verified note map are not enough. A displayed A must
1141 point to an actual A in the score.
- 1142 8. **Score alignment.** For known repertoire, the played note sequence must align to a specific
1143 location in the score. The match can ignore rhythm during the first pitch-class search, but
1144 accepted score display needs exact score-note identity and a stable location.
- 1145 9. **Score-free exercise handling.** If Alan invents an exercise or practices a technique pattern that
1146 is not in the score, Curtis should still transcribe and group the pattern. It should not force the
1147 pattern into Scherzo-Tarantelle, Haydn, or another piece.
- 1148 10. **Repeat grouping.** Repeated attempts of the same passage are grouped as attempts, loops,
1149 restarts, or slow/faster passes. The UI should not dump the same measures repeatedly.
- 1150 11. **Heat maps.** Practice density, repetition density, problem density, and improvement layers appear
1151 on actual score coordinates when a score match exists, or on the generated pattern/transcription
1152 when the material is score-free.
- 1153 12. **Curtis-level observations.** Observations are specific and evidence-tied: what happened, where
1154 it happened, how often it happened, whether it improved, and why the issue blocks Curtis-level
1155 readiness. Generic advice is not accepted.
- 1156 13. **Compact interface.** Expanded daily entries stay compact. The useful unit is the grouped
1157 score–transcription–audio–video pair, not a large dashboard or a long explanatory panel.
- 1158 14. **Regression protection.** Wrong score-note boxes, failed transcriptions, unmatched notation,
1159 stale piece labels, and non-playing practice-time credit cannot silently reappear after a later
1160 change.

5.3 Acceptance gates

The following gates define whether progress is real. If a change fails one of these gates, it is not counted as solved, even if the UI looks better.

Archive gate.

The strict practice ledger must list every video from violin 1 forward that is either violin-numbered or date-titled as practice. Same-day videos must merge into one daily record. Each record stores source video id, title, uploaded date, inferred practice date, uploaded duration, and processing coverage.

Practice-time gate.

Active violin time is the sum of detected playing intervals, not the uploaded duration. The API must expose measured active time, checked video time, unmeasured video time, and estimate status separately. The UI label should use total practice time for active violin time and uploaded video for total source duration.

Evidence-media gate.

A transcription line is accepted only if the exact audio and video window are available locally, playable in the same Curtis page, and time-linked to the source record.

Note gate.

A displayed note must agree with the paired audio. The note label, octave where claimed, and staff position must be internally consistent. If the audio says A4, the rendered staff note must be A4, not F, G, B, or a visually adjacent note.

Score gate.

An accepted score snippet must come from the actual piece source: a verified source-score crop, verified MusicXML engraving export, or manually verified note-coordinate map for the same reference range. Generated notation may show the transcription, but it cannot appear as the accepted score snippet.

Single-note gate.

A one-note score display is allowed only as a calibration anchor, and only if both sides explicitly show the same pitch name and the score-side coordinate is verified. It must not be used as proof of a long passage.

Phrase gate.

A real phrase match requires a contiguous pitch-class sequence in the score or pattern source. Rhythm can be ignored for early matching, but accepted phrase display must keep the exact played sequence, the exact score sequence, and any omitted or uncertain notes.

Notation gate.

Hand-built notation SVG is not the long-term display path. The production path must use a professional renderer or renderer-equivalent layout with verified clef, key, staff, note, rest, spacing, and responsive behavior.

Exercise gate.

If no score match exists and the material forms a repeated exercise, the system must label it as an exercise or pattern, preserve the generated transcription, and avoid adding a fake repertoire entry.

Observation gate.

A Curtis-level observation must cite the clip, transcription event, score measure or pattern segment, issue type, repeat count, direction of change, and readiness consequence.

UI gate.

The day row stays chronological, the expanded state is compact, and only grouped evidence appears by default. Raw untrusted transcription stays hidden or goes to the transcription-run PDF with clear status.

Paper gate.

Any meaningful progress changes this section, the system-state table, and the limitation text in the same work cycle.

5.4 Required source material

1211

The implementation cannot be completed by code alone. It needs a source bundle that makes the audio, score, and corrections checkable. The Wieniawski Scherzo-Tarantelle solo PDF is now present locally as the first source-score asset, and its first verified symbolic map covers the nine-note opening phrase only.

1215

1. **Raw or locally captured practice media.** Curtis needs stable local access to audio/video windows from the YouTube archive or owner-provided files. Metadata alone cannot produce practice time, clips, or transcription.
2. **Source-confirmed piece names by day.** Alan can supply piece names when automatic detection is uncertain. These labels should seed matching but never become score evidence by themselves.
3. **Symbolic scores for known repertoire.** For each piece, Curtis needs MusicXML, MEI, a verified MuseScore export, or an equivalent symbolic note map with measure numbers, pitches, durations, voices, and staff positions.
4. **Score images tied to symbolic coordinates.** To show actual score snippets, the image/PDF rendering must be linked to the same symbolic note ids or verified coordinate boxes.
5. **Manual corrections.** Alan or a reviewer must be able to correct a note, octave, onset, rhythm, score location, or false match. These corrections become training and regression data.
6. **Calibration recordings.** Labeled scales, arpeggios, open strings, repeated notes, shifts, slow exercises, and fast exercises help tune pitch detection before attempting long phrases.
7. **Gold benchmark clips.** A small set of manually checked audio–transcription–score pairs is required before the app can claim improvement. The benchmark must include easy anchors, repeated notes, fast runs, arpeggios, and real Scherzo-Tarantelle fragments.

1216

1217

1218

1219

1220

1221

1222

1223

1224

1225

1226

1227

1228

1229

1230

1231

1232

5.5 Implementation plan

1233

Phase 0: Make the paper the tracker

1234

Status: complete in this revision. The first required step is to make the plan durable in the Curtis paper PDF. The paper now records the definition of solved, the current baseline, the gates, the phase plan, and the update protocol. Future changes are not complete unless the paper is checked or updated.

1235

1236

1237

Phase 1: Build the canonical evidence ledger

The backend needs a stricter data model for practice evidence. The current modules already include the daily-record builder, media sampler, transcription engine, symbolic-score parser, score-asset layer, reference corpus, and correction layer. The next version should make the following entities explicit:

- **source_video**: immutable source id, title, URL, uploaded time, duration, practice-date basis, and ledger inclusion basis.
- **practice_day**: date, list of source videos, uploaded duration, active-practice coverage, piece labels, and status.
- **active_interval**: source id, start, end, confidence, detector version, reason for inclusion, and reason for exclusion when withheld.
- **media_clip**: local video path, local audio path, waveform path, source interval, duration, and playable route.
- **note_event**: onset, offset, pitch class, octave, cents deviation, confidence, detector sources, uncertainty reason, and correction state.
- **score_note**: piece id, movement, measure, voice, pitch class, octave, duration, note id, and rendered coordinates.
- **match_candidate**: played note ids, score note ids or exercise-pattern ids, match score, mismatch list, and display eligibility.
- **accepted_evidence**: only the groups that pass the gates; these are the only groups the main UI can show.
- **rejected_evidence**: wrong notes, wrong score boxes, false labels, failed transcriptions, and model guesses that must not reappear.

The acceptance condition for this phase is that a wrong A-to-B or A-to-G score crop can be stored as rejected evidence and cannot be selected by the display builder again.

Phase 2: Complete the full active-practice scan

1262

The practice-time problem is separate from transcription. Curtis should first scan the full 234 h 57 min uploaded practice ledger for violin-playing intervals with a low-cost pass. The pass should use audio energy, violin-range voiced pitch, onset density, spectral shape, and visual fallback only where necessary. It should not require full transcription.

1263

1264

1265

1266

The fast pass should walk every practice video in chronological order and store approximate active intervals. A second exact pass should refine boundaries around detected regions. The UI should show three numbers for every day: uploaded video time, checked video time, and total practice time. While scanning is partial, the estimate stays labeled as estimated. When every video has been scanned, the estimate should be replaced by measured total practice time.

1267

1268

1269

1270

1271

Acceptance for this phase:

1272

- every strict-ledger video has a coverage record;
- every day has uploaded duration and active-practice status;
- non-playing windows are excluded;
- active-practice total can be recomputed from stored intervals;
- the paper table is updated with checked duration, active duration, unmeasured duration, and estimate status.

1273

1274

1275

1276

1277

1278

Phase 3: Generate stable local evidence clips

1279

Every accepted interval and every candidate transcription window needs local media. Curtis should generate short MP4 clips and audio clips with stable IDs derived from source video id and time range. The browser should play the local clip and audio without YouTube embedding. The audio used for transcription must be the same audio the user can play.

1280

1281

1282

1283

Acceptance for this phase:

1284

- each accepted transcription event links to local video and local audio;
- the site can play the clip in the same window;
- source time, local file, and rendered transcription agree;
- dead local media routes fail the evidence gate rather than rendering a broken accepted group.

1285

1286

1287

1288

Phase 4: Build the verified score library

The current weak link is score-side truth. Curtis must stop relying on guessed scanned-score boxes. For Wieniawski Scherzo-Tarantelle and any future piece, the score library needs symbolic notes and rendered coordinates.

The best path is:

1. acquire a legal public-domain score image or PDF;
2. obtain or create a symbolic MusicXML/MEI/MuseScore representation;
3. render the symbolic score to image or SVG with stable note ids;
4. map each rendered note to pitch, octave, measure, and coordinates;
5. manually verify a representative set of anchors, especially A, D, G, B, fast runs, ledger lines, accidentals, and octave-displaced passages;
6. reject any score source where the note map cannot distinguish adjacent notes reliably.

Acceptance for this phase:

- a score-side A in the UI is generated from an actual score-side A;
- the rendered crop includes enough context to be legible;
- the crop does not cut off clefs, notes, beams, stems, or measure context;
- single-note anchors are visually verified before display;
- phrase snippets use the exact matched symbolic notes, not a guessed image region.

Phase 5: Replace fragile note extraction with a multi-stage transcription engine

The note detector needs to work on violin practice, not only on clean isolated notes. The engine should process active intervals in stages:

1. normalize audio and remove long silence;
2. isolate violin-dominant energy where possible;
3. detect onsets and repeated attacks before pitch smoothing;

4. run multiple pitch estimators over the same window; 1313
5. separate stable pitch, glissando/shift, vibrato, noise, and uncertain frames; 1314
6. group frames into note events by onset, pitch stability, and duration; 1315
7. preserve fast pitch changes instead of smoothing them into one repeated pitch; 1316
8. estimate rests and pauses between attempts; 1317
9. detect double stops or chords as multi-note events when monophonic assumptions fail; 1318
10. attach confidence and uncertainty to every event; 1319
11. store rejected alternatives so repeated failures can be debugged. 1320

Acceptance for this phase: 1321

- open strings and simple scales transcribe correctly; 1322
- repeated notes remain repeated notes with distinct onsets; 1323
- arpeggios become changing pitch sequences, not a repeated-note collapse; 1324
- obvious fast runs preserve the contour and most pitch classes; 1325
- uncertain events are labeled uncertain rather than shown as clean notation; 1326
- every displayed note can be played from its paired audio. 1327

Phase 6: Make transcription score-aware 1328

For known repertoire, Curtis should not transcribe in isolation and then guess the piece later. The user-supplied or detected piece name should provide a score prior. The aligner should search the score for pitch-class sequences that match the audio-derived note stream, allowing restarts, skipped notes, slow practice, repeated loops, tempo changes, and rhythm mismatch during early search. 1329 1330 1331 1332

The alignment should use multiple layers: 1333

- pitch-class subsequence search for easy anchors; 1334
- interval-contour search when absolute pitch is uncertain; 1335
- dynamic time warping for variable tempo; 1336

- Viterbi-style path selection through the symbolic score for longer passages;
- restart and loop detection when the path jumps backward;
- score-free fallback when no score path is plausible.

Acceptance for this phase:

- one-note anchors pass only when the score coordinate is verified;
- three-to-five-note sequences can be matched to exact score notes;
- longer phrases show played notes beside matched score notes;
- mismatches are visible rather than hidden;
- rhythm mismatch can be tolerated during matching but not during final notation labeling;
- accepted score matches survive a musician spot-check.

Phase 7: Support score-free technique exercises

Alan may invent an exercise during practice. Curtis must treat this as first-class evidence. If the note stream is coherent but does not match the named score, the system should cluster repeated patterns, render generated notation, and label the material as a technique pattern or exercise.

Acceptance for this phase:

- repeated made-up patterns group together;
- the generated transcription is linked to audio/video;
- no fake score snippet appears;
- the repertoire entry records technique-pattern evidence separately from piece evidence;
- heat maps can show pattern density without pretending there is a score location.

Phase 8: Add manual correction and training

Fully automatic transcription will not become reliable without corrections. Curtis needs a correction UI that allows Alan to say: this note is A, this staff note is wrong, this score box points to B, this clip is non-playing, this rhythm is wrong, this is Scherzo-Tarantelle, this is a made-up exercise, or this score match is false.

Status: partially implemented. The gold review loop now covers audio-phrase acceptance/rejection and score-phrase acceptance/rejection from queued detected clips. It is intentionally narrow: it can create accepted audio truth and exact score-ready truth, but it does not yet provide detailed onset/rhythm editing, score-box coordinate editing, batch keyboard review, waveform overlays, or full measure-level correction.

Every correction should become data:

- corrected note events become gold labels;
- rejected matches become negative examples;
- accepted matches become benchmark fixtures;
- corrected piece names update source labels without unlocking score snippets by themselves;
- corrections invalidate stale derived evidence.

Acceptance for this phase:

- Alan can accept or reject queued note windows without editing code;
- accepted and rejected examples appear in the gold/truth set;
- the same failure cannot reappear after rerendering;
- the PDF tracker records the correction count and benchmark count.

Phase 9: Build the benchmark suite

Curtis needs objective tests before claiming improvement. The benchmark should include:

- open-string clips;
- G, D, A, and E major/minor scales where available;
- arpeggios;
- repeated single-note attacks;
- fast scalar runs;
- shifts with sliding or unstable pitch;
- Scherzo-Tarantelle fragments with verified score locations;

- Haydn fragments with verified score locations;
- score-free exercises;
- non-playing windows that must not count as practice time.

Metrics should include note precision, note recall, pitch-class accuracy, octave accuracy when claimed, onset error, duration error, false-note rate, repeated-note recall, phrase-match precision, phrase-match recall, score-coordinate accuracy, false accepted score match count, active-practice precision, active-practice recall, and UI evidence completeness.

Acceptance for this phase:

- tests fail on the known A-to-wrong-score-note examples;
- tests fail on repeated-pitch collapse;
- tests fail when accepted notation lacks paired audio/video;
- tests fail when practice time includes non-playing footage;
- tests pass only when the displayed evidence is actually trustworthy.

Phase 10: Use professional notation rendering

The current custom notation path is only a temporary evidence display. The solved system should render notation from MusicXML, MEI, or another structured notation representation through a professional renderer or renderer-equivalent layout. The renderer must handle treble clef placement, staff scale, ledger lines, rests, stems, beams, ties, slurs where known, key signatures, accidentals, note spacing, measure widths, and responsive crop behavior.

Acceptance for this phase:

- treble clef is correctly centered around the G-line spiral;
- the staff note for A4 appears in the A4 position;
- key signature and accidentals follow the chosen notation policy;
- snippets look publication-quality at desktop and mobile widths;
- the same notation source can export to the transcription-run PDF.

Phase 11: Build heat maps from matched coordinates

1412

Heat maps should be computed from actual practice behavior. For score-matched passages, each matched score note or measure receives density values. For score-free exercises, each pattern segment receives density values.

1413

1414

1415

The layers are:

1416

- practice density: active time spent on the passage;
- repetition density: number of attempts or loops;
- problem density: restarts, stops, slowdowns, unstable notes, and failed attempts;
- improvement: later attempts cleaner than earlier attempts.

1417

1418

1419

1420

Acceptance for this phase:

1421

- a heat-map mark can be traced to clips and note events;
- a score-free exercise can still have a pattern heat map;
- heat maps do not appear as decorative overlays when no match exists;
- the day-level heat map and repertoire heat map use the same evidence.

1422

1423

1424

1425

Phase 12: Generate specific Curtis-level observations

1426

Observation generation should happen after transcription and alignment, not before. The observation engine should compare attempts of the same passage and classify issues as pitch, rhythm, bow, shift, tone, articulation, tempo, consistency, musical shape, or visible tension.

1427

1428

1429

The output for each main passage should include:

1430

- likely passage or exercise pattern;
- evidence clip;
- transcription snippet;
- score snippet when verified;
- exact issue;
- number of attempts where it occurred;

1431

1432

1433

1434

1435

1436

- 1437 • whether it improved, stayed the same, or got worse;
- 1438 • why it blocks Curtis-level readiness.

1439 Acceptance for this phase:

- 1440 • observations cite evidence;
- 1441 • generic advice is absent;
- 1442 • a day ends with one specific main blocker;
- 1443 • rejected or uncertain transcription does not feed readiness claims.

1444 **Phase 13: Keep the UI compact**

1445 The main page should show the paper card, analyzed practice days, and repertoire. Expanded daily
1446 entries should show one or a small number of grouped evidence pairs, not a full debug dashboard. The
1447 day marker should remain visible on the left so boundaries between days are clear. Videos should be
1448 small enough that a day entry fits on screen when possible.

1449 Acceptance for this phase:

- 1450 • one daily group contains score, clip, audio, transcription, and the minimum useful status;
- 1451 • raw text explanations are removed from the main view;
- 1452 • chronological order is stable;
- 1453 • desktop and mobile render without overlap, excessive height, or broken media controls;
- 1454 • unresolved material is visible as state, not as a fake dashboard.

1455 **Phase 14: Automate paper updates**

1456 The paper is now the progress tracker. Future meaningful Curtis changes should update:

- 1457 • live state numbers;
- 1458 • solved gates;
- 1459 • benchmark metrics;
- 1460 • accepted score-match count;
- 1461 • accepted transcription duration;

- full archive scan coverage; 1462
- remaining blockers; 1463
- limitations. 1464

Acceptance for this phase is procedural: after every Curtis change that affects transcription, score matching, practice time, clips, heat maps, repertoire, observations, or evidence gates, the LaTeX paper is checked, rebuilt, and the root `paper.pdf` is refreshed. 1465 1466 1467

5.6 Milestones 1468

5.7 Immediate next implementation 1469

After this PDF tracker, the first concrete implementation target is **full active-practice coverage plus benchmark/correction scaffolding**. This is the first useful code milestone because transcription cannot be trusted without knowing which source windows contain violin playing, and it cannot improve reliably without a gold correction loop. The first gold review loop is now implemented for queued audio/score phrase labels, but the larger benchmark suite and full archive coverage remain incomplete. The 2026-05-13 11:57 AM EDT backend update completed the coverage and correction scaffolding portion: 1470 1471 1472 1473 1474 1475 1476

- added a canonical active-practice coverage ledger exposed through the review API and `/api/curtis/active-practice-coverage`; 1477 1478
- separated uploaded video duration, checked sampled video duration, measured active practice time, candidate violin windows, unmeasured video time, and estimate state; 1479 1480
- counted checked non-playing sampled windows as checked video without adding them to practice time; 1481 1482
- added `/api/curtis/evidence-corrections` for note, score, match, piece, and active-interval corrections; 1483 1484
- added benchmark fixtures for rejected score-note and match corrections; 1485
- added regression tests for checked non-playing windows, wrong score-note benchmark creation, and rejecting accepted score evidence when the displayed score note conflicts with the observed note. 1486 1487 1488

Table 4. Milestone plan for moving from current evidence gates to solved long-phrase transcription.

Milestone	Target	Acceptance evidence	Status
M0	Paper tracker	Roadmap is included in the Curtis paper PDF and becomes the update record.	Complete in this revision.
M1	Practice time	All 234 h 57 min of strict-ledger video has active-practice coverage and per-day totals.	Active-scan route deployed; live state now has 331 active intervals from 106 sample results, 2 h 28 min checked video, 2 h 21 min measured active practice, and 250 queued windows; full chronological scan still pending.
M2	Score truth	Scherzo-Tarantelle has verified symbolic notes and score coordinates.	Advanced; nine opening notes, one Staff 4 six-note accepted source/audio window, and one adjacent Staff 4 seven-note source-only extension are verified in MusicXML. Broader score coordinates remain pending.
M3	One-note anchor	A displayed A from audio is matched to a verified score-side A, with no wrong-note crop.	Not accepted yet; audio-only D/A fragments are useful, but all A-specific score anchors remain rejected until the score-side notehead and octave are verified.
M4	Short sequence	Three-to-five played notes align to exact score notes and render with local audio/video.	Advanced past the short-sequence threshold; one May 3 six-note audio phrase matches the verified Staff 4 source window by exact MIDI. Long phrases remain pending.
M5	Long phrase	A 10–30 s passage transcription matches the paired audio and score location.	Pending.
M6	Repeat grouping	Multiple attempts of the same passage group into loops, restarts, slow attempts, and faster attempts.	Pending.
M7	Heat map	Practice density and problem density appear on verified score coordinates.	Pending.
M8	Repertoire update	Piece entries update from accepted daily evidence and show active time, clips, snippets, and blockers.	Pending.
M9	Readiness observation	Main blocker is generated from accepted evidence, not generic advice.	Pending.
M10	Regression lock	Tests prevent failed notation, wrong score boxes, and non-playing practice-time credit from returning.	Advanced; exact-MIDI manifest verifies three positive source phrases, accepts one live source/audio phrase, blocks four false phrase maps, records the audited right-2 Staff 4 expansion as a rejected regression, moves currentBest to the right-1 window, mines stored May 3 audio windows for exact right-1/right-2 MIDI candidates, and now rescans the original Staff 4 source audio before another expansion can be reviewed.

The 2026-05-13 12:33 PM EDT backend update completed the active-scan route portion: 1489

- added a persistent `activePracticeScan` state record with active intervals, sample results, pending windows, and run history; 1490
1491
- scans stored local media samples only after the violin-presence gate is positive; 1492
- extracts sound-active sub-intervals inside the source window and stores them as active-practice intervals; 1493
1494
- records negative violin-presence samples as checked without adding practice time; 1495
- feeds persisted active intervals into per-day totals and the active-practice coverage ledger; 1496
- exposes active-scan status and run endpoints so the deployed runtime can perform the pass without transcription; 1497
1498
- queues the next unchecked 90-second strict-ledger windows for owner media capture. 1499

The 2026-05-13 12:59 PM EDT deployed scan update completed the first live run and exposed a practice-time accounting failure mode: sparse, captured samples can make an archive-wide estimate look too certain. The implementation now records active-scan status counts, caps active seconds by checked source coverage, and treats active-scan results as the practice-time source for scanned samples instead of letting noisy transcription durations override them. 1500
1501
1502
1503
1504

The 2026-05-13 03:26 PM EDT queue update connected the active-scan queue to the owner-media sync path. The sync path now asks the deployed active-scan run endpoint for pending windows, keeps the response compact, and places those windows ahead of generic expansion or blind sampling. The pending queue remains the source of work for M1: each captured window must still be downloaded or browser-captured, locally classified for violin presence, uploaded, rescanned, and recorded before it becomes checked practice-time coverage. 1505
1506
1507
1508
1509
1510

The 2026-05-13 03:37 PM EDT owner-sync run proved that the local upload/rescan loop still works after the queue update. It uploaded one cached violin-positive 5/3 window and the deployed active scan converted it into one additional active-practice interval. Because cached violin-positive media is still prioritized ahead of new capture, this run did not consume the first chronological pending window. The next M1 step should either exhaust the cached positive backlog first or run a queue-only owner-sync mode so the pending list starting at `violin 1` advances. 1511
1512
1513
1514
1515
1516

The 2026-05-13 04:05 PM EDT update added that queue-only owner-sync mode and used it to consume the first chronological pending window. The violin 1 *0-90 window was captured, classified as non-playing/unclear, uploaded, and folded into the scan results as checked non-violin evidence. This is correct progress because it prevents setup or non-playing footage from being counted as violin practice. It does not advance transcription because no violin-positive musical material was present in that window. The UI now shows the current long-phrase transcription completion percentage and the complete done/open gate list at the bottom of the Curtis page so the remaining work is visible without turning daily entries into a large debug dashboard.

The 2026-05-13 05:44 PM EDT update advanced the same queue from *90-180 through *450-540. Those windows produced active-practice evidence and moved checked coverage to 2 h 3 min. The timeout exposed a resume failure: cached active-scan pending windows could be captured but then skipped on the next queue-only run. The helper now includes matching cached pending-window files in queue-only order, including non-playing cached windows, so an interrupted scan can resume rather than forcing Alan or Codex to manage the cache manually. The coverage ledger also now treats unchecked transcription-only remnants as zero practice time; active practice cannot exceed checked video coverage.

The 2026-05-13 06:01 PM EDT update consumed the next chronological queue window, *540-630. That window was violin-positive and added four active intervals after the active-practice scan. The run succeeded without a timeout at batch size one, and the next pending window became *630-720. This kept the current work focused on M1 coverage rather than score matching or long-phrase notation.

The 2026-05-13 06:26 PM EDT update consumed the next chronological queue window, *630-720. That window was violin-positive and added five active intervals after the active-practice scan. The command wrapper timed out while the owner-sync process was still running, so the process was allowed to finish and then verified from live state instead of starting a duplicate run. The next pending window is now *720-810.

The 2026-05-13 07:55 PM EDT update consumed the next chronological queue window, *720-810. That window was violin-positive and added three active intervals after the active-practice scan. The owner-sync process completed cleanly with a longer timeout, so no duplicate capture or manual cache recovery was needed. The next pending window is now *810-900.

The 2026-05-13 08:41 PM EDT status-surface update made the Curtis page itself carry the imple-

mentation state instead of leaving the percentage and plan in the chat thread. The bottom tracker is now labeled as long-phrase transcription implementation status and is driven by the same backend gate object as the API. The visible panel reports 29 percent completion, 2 h 7 min checked video out of 213 h 23 min, 2 h measured active practice, 0 exact score windows, 0 accepted long phrases, 250 queued windows, and 250 active intervals. It also exposes a compact done list, an open-work list, an eight-phase implementation plan, and a collapsed detailed gate checklist. This is a presentation and tracking improvement, not a transcription breakthrough: score-truth verification, phrase-level note/rhythm extraction, exact score alignment, heat maps, and Curtis-level observations remain open.

The 2026-05-13 09:25 PM EDT implementation pass advanced the M1 scan and M10 benchmark loop. The queue-only sync consumed violin 1 windows *810-900 and *900-990; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31 percent implementation completion, 2 h 10 min checked video out of 213 h 23 min, 2 h 3 min measured active practice, 92 sample results, 82 active-violin samples, 10 checked non-violin samples, 257 active intervals, and 250 queued windows. The correction ledger now contains three benchmark fixtures for real wrong-score-note failures: A4/B-crop, A4/G4-crop, and A5/G5-crop. Those cases are progress only because they prevent the same bad score matches from being counted as solved evidence. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 10:05 PM EDT implementation pass continued the M1 scan and fixed a tracker presentation gap. The queue-only sync consumed violin 1 windows *990-1080 and *1080-1170; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31.13 percent exact weighted implementation completion, 2 h 13 min checked video out of 213 h 23 min, 2 h 6 min measured active practice, 94 sample results, 84 active-violin samples, 10 checked non-violin samples, 269 active intervals, and 250 queued windows. The rounded public label still reads 31 percent, but the API and site now expose exact weighted progress so small M1 coverage gains do not disappear. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 11:14 PM EDT implementation pass continued M1 and tightened the progress calculation itself. The queue-only sync consumed violin 1 windows *1170-1260 and *1260-1350; the first became violin-positive source evidence and the second became checked non-playing/unclear evidence. The deployed active-practice scan folded both into the coverage ledger. Live state now

reports 96 sample results, 85 active-violin samples, 11 checked non-violin samples, 283 active intervals, 2 h 16 min checked video, 2 h 7 min measured active practice, and a 199 h 19 min provisional archive estimate. The tracker implementation now sums unrounded gate points before formatting the public label, so this small but real coverage gain moves the expected exact implementation label to 31.128 percent after deployment. Curtis still has 0 accepted exact score windows and 0 accepted long phrases. The 2026-05-13 11:38 PM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *1350-1440 and *1440-1530; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31.131 percent exact weighted implementation completion, 2 h 19 min checked video out of 213 h 23 min, 2 h 10 min measured active practice, 98 sample results, 87 active-violin samples, 11 checked non-violin samples, 294 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 11:55 PM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *1530-1620 and *1620-1710; the first became checked non-playing/unclear evidence and the second became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.134 percent exact weighted implementation completion, 2 h 22 min checked video out of 213 h 23 min, 2 h 12 min measured active practice, 100 sample results, 88 active-violin samples, 12 checked non-violin samples, 300 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 12:09 AM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *1710-1800 and *1800-1890; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.136 percent exact weighted implementation completion, 2 h 25 min checked video out of 213 h 23 min, 2 h 15 min measured active practice, 102 sample results, 90 active-violin samples, 12 checked non-violin samples, 313 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 12:35 AM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *1890-1980 and *1980-2070; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.139 percent exact weighted implementation completion, 2 h 28 min checked video out of 213 h 23 min, 2 h

18 min measured active practice, 104 sample results, 92 active-violin samples, 12 checked non-violin samples, 322 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 03:00 AM EDT implementation pass continued M1. The queue-only sync consumed violin 1 windows *2070-2160 and *2160-2250; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.142 percent exact weighted implementation completion, 2 h 31 min checked video out of 213 h 23 min, 2 h 21 min measured active practice, 106 sample results, 94 active-violin samples, 12 checked non-violin samples, 331 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 05:11 AM EDT source-score acceleration pass advanced M2 and M4 instead of continuing only the slow archive scan. The implementation added a local MusicXML source excerpt for the Scherzo-Tarantelle solo-violin opening motif, wired it into the Wieniawski score target, and changed the symbolic matcher so this source target can accept a four-note measure-level run with at least three distinct pitch classes. The parser now preserves written flats such as B-flat for staff placement while still matching their pitch class to detected audio, which closes the failure mode where a normalized A-sharp could render on the wrong staff position. The score snippet renderer now receives the target key signature and renders a G-minor two-flat key signature before the matched notes. The completion tracker now separates `acceptedMeasureMatchCount` from `longPhraseAcceptedCount`, so Curtis can show one verified measure-level score/audio match without claiming solved long-phrase transcription. The regression suite increased to 103 tests and now covers source-backed Wieniawski symbolic notes, four-note source-measure matching, written-flat staff placement, accepted-measure counting, and the continued rejection of one-note visual crops. This pass does not solve long-phrase transcription; it creates the first narrow, test-covered lane from source score notes to a grouped score–audio–video–transcription match.

The 2026-05-16 10:30 PM EDT source-review pass completed the next narrow lane that the roadmap called for. One queued Staff 4 review packet from the local IMSLP Scherzo-Tarantelle solo part was converted into verified MusicXML for the visible E-flat5–E-flat5–C5–E-flat5–E-flat5 source measure-window. The accepted crop is a non-generated source-score image from page 2 with boxes centered on the five noteheads. The live May 3 detected phrase D#5 D#5 C5 D#5 D#5 is accepted against that source window because the MIDI sequence 75 75 72 75 75 matches the verified score

exactly; the displayed score spelling remains Eb5 Eb5 C5 Eb5 Eb5. The truth manifest now verifies two source-positive exact-MIDI phrases, marks one as live accepted, and keeps the three known false phrase maps blocked. The regression subset increased this lane to 34 focused tests covering the MusicXML slice, source-snippet gate, exact-MIDI acceptance, written-flat display, source-range verification, and rejected phrase regressions. This is still not long-phrase transcription: it is one verified source/audio phrase that can be extended outward into adjacent measures.

The 2026-05-17 anchored-expansion harness implements that outward-extension step as a gate rather than a manual judgment. The solve tracker now starts from the accepted Staff 4 source/audio anchor and tests adjacent left and right source-note ranges against available audio-agreed note runs. Current result after the right-1 promotion: the six-note lane Eb5 Eb5 C5 Eb5 Eb5 Eb5 is accepted, and the seven-note right-side lane Eb5 Eb5 C5 Eb5 Eb5 Eb5 C5 remains the next audit target. This prevents Curtis from treating a new random phrase candidate as progress while the current accepted lane has not yet extended cleanly.

The next 2026-05-17 expansion-search pass completes the first search-pool improvement requested by that gate. The expansion harness now searches raw detected note series from the daily record in addition to already-ranked score-candidate cards, records whether the best run came from the raw series or ranked group pool, reports raw-run counts in the tracker, and has a regression test proving that a raw exact seven-note Staff 4 run becomes `ready_for_truth_review` rather than being missed. This still does not accept a longer phrase automatically; the expanded phrase remains hidden until exact MIDI, paired audio, source crop, and truth evidence all pass.

The 2026-05-17 audit-rejection pass turns the audited right-2 failure into a permanent negative example instead of leaving it as the next task. The manifest blocks four false phrase maps, including the Staff 4 source/audio expansion whose sixth note is D5 against source Eb5. The expansion harness keeps that rejected case visible in the audit items but removes it from the current-best path. A later right-1 exact audio candidate for Eb5 Eb5 C5 Eb5 Eb5 Eb5 now passes and is accepted; the useful constraint remains that the old seven-note expansion cannot be recycled as a solved phrase.

The 2026-05-17 adjacent-mining pass implements the next search step directly. Rather than waiting for a score candidate to happen to surface, Curtis now asks a narrow question over stored May 3 detected audio: does any window contain 75 75 72 75 75 75 or 75 75 72 75 75 75 72? If yes, the window is exposed as an exact mining candidate for audit. If no, the tracker records the nearest stored miss and tells the next sync/transcription pass to mine more neighboring active windows. This

keeps the accepted Staff 4 lane fixed and prevents the system from treating unrelated score matches as progress.

The 2026-05-17 03:23 AM EDT source-audio rescanner removes one bottleneck in that search. Curtis no longer depends only on note windows that happened to be stored before the current extractor and score-lane gates existed. It can now reopen the actual source media for the accepted Staff 4 anchor, extract a padded local audio window, run the current transcription pipeline over that audio, and merge the resulting note runs into the exact-MIDI expansion and mining harnesses. A rescanned source run is search evidence, not accepted evidence: if it disagrees with the verified source lane, it is reported as a blocker; if it agrees, it becomes an audit candidate, not a visible match.

The 2026-05-17 anchor-persistence pass prevents a fragile live rollback. The accepted Staff 4 anchor is now a truth-manifest object, not only a side effect of the current daily matchGroups display. If the display layer has zero accepted groups for May 3, the backend still resolves the verified source/audio anchor, uses its stored sample and timing to run source-audio rescanning, and feeds the same anchor into expansion and adjacent mining. The duplicate guard prevents double-counting when both the visible match group and the truth-manifest anchor are present.

The 2026-05-17 12:56 PM EDT tiled-rescan pass is the next implementation step from that state. The source-audio rescanner now covers more of the same accepted source sample by extracting five overlapping windows, not one. This increases the chance of finding the adjacent Eb5 continuation in the original audio while preserving the exact-MIDI gate and keeping every mismatched tile candidate-only.

The 2026-05-17 01:43 PM EDT anchor-guided reproduction pass fixes the current-detector rollback exposed by the tiled scan. The broad transcription pass can still miss or segment the accepted Staff 4 anchor badly, but the rescanner now has a separate seeded check: use the accepted note-window timings, rerun current pYIN/YIN/spectral-onset agreement on those exact audio slices, and mark the anchor reproduced only when the exact MIDI sequence 75 75 72 75 75 is confirmed. Live verification reported five guided anchor events and reproduced anchor status, while the adjacent exact-MIDI search still found zero right-extension candidates after 1671 windows. The notation renderer was also tightened so accidental spelling affects display: A# can render as Bb in a flat-key context or with a local accidental, rather than putting a natural A notehead on the staff. The 2026-05-17 02:19 PM EDT renderer patch moves local sharps and flats from text entities into drawn SVG accidental glyphs, keeps high-note accidentals inside the visible staff viewport, and verifies that A#, D#, Bb, Eb, and the Bb/Eb key signature all render with matching visible symbols.

The 2026-05-17 04:49 PM EDT adjacent-guided phrase pass is the next concrete implementation step from that state. Instead of only widening source windows again, Curtis now derives the Staff 4 right-1 and right-2 MIDI targets from the accepted source lane, infers the next note windows from the accepted anchor timing, and checks each expected note with current pYIN/YIN/spectral exact-MIDI votes. If every note reproduces, the longest adjacent target enters the hidden exact-MIDI mining pool as `staff4_adjacent_guided_current_detector`; if any note fails, the rescan record stores the first failed note, expected MIDI, detector votes, and seed window. The regression suite now proves that a broad transcription miss can still produce a seven-note adjacent phrase when current audio agrees with every source MIDI value, without allowing that phrase to bypass source crop and truth gates.

The 2026-05-17 05:29 PM EDT adjacent-onset sweep pass follows the same fixed-lane rule but removes a brittle timing assumption. Inferred continuation notes now sweep bounded early and late offsets around the predicted onset before failing. The accepted five anchor notes remain fixed; only notes beyond the anchor are swept. Each attempt records the offset, note window, exact detector sources, full detector votes, and best failed attempt. A new regression proves that when the base inferred sixth note returns D5 but a nearby offset returns source Eb5, the seven-note Staff 4 target can enter the hidden exact-MIDI mining pool with the nonzero timing offset recorded.

The 2026-05-17 05:54 PM EDT failure-localization and notation-quality pass completes the next diagnostic layer. The source-audio rescanner now carries a compact first-failure object through the completion API and site tracker: direction, target sequence, failed note index, expected note and MIDI, best swept offset, detector consensus MIDI, and failure kind. This prevents the next pass from asking the broad question “why no phrase?” and instead asks the narrow question “why did this exact Staff 4 adjacent note fail?” The notation renderer also received a global symbol pass. Sharps, flats, key-signature flats, the treble clef, high accidentals, note labels, and CSS stroke/fill behavior are now covered by regression tests so accepted snippets cannot silently regress into crude or musically misleading glyphs.

The 2026-05-17 06:14 PM EDT notation visual-QA pass fixed the remaining symbol-rendering failure caught in browser review. The clef itself was no longer a note-name error, but the staff viewport still clipped the bottom of the glyph and compact responsive panels could shrink the notation into a lower-quality mobile rendering. Curtis now uses a taller notation viewport for both accepted transcription staves and notation gates, keeps compact matched panels from squeezing the clef, and preserves the same sharp/flat stroke and fill geometry across responsive overrides. Browser screenshots

at desktop and mobile widths verify that the treble clef is not clipped, local sharps remain visible as sharps, local flats remain visible as flats, and the two-flat key signature renders on B-flat/E-flat staff positions.

The 2026-05-17 08:40 PM EDT Staff 4 first-failure audit gate adds the review loop needed before the lane can move again. The first failed adjacent-note object now preserves source spelling separately from detector spelling, so a Staff 4 source Eb5 is no longer surfaced as D#5 just because the detector uses sharp names internally. The audit packet now records a decision: exact accepted source/audio evidence may extend the lane, wrong-MIDI detector consensus becomes a rejected regression case, and missing or non-reproduced detector evidence stays queued for gold review or rescanning. The site shows the decision label and source/audio note pair, which keeps the current work centered on the real question: whether the next adjacent Staff 4 source note is audibly and score-truth verified.

The 2026-05-17 10:46 PM EDT Staff 4 right-1 promotion converts the current exact audio candidate into accepted source/audio truth. The accepted phrase is source Eb5 Eb5 C5 Eb5 Eb5 Eb5, detected audio D#5 D#5 C5 D#5 D#5 D#5, exact MIDI 75 75 72 75 75 75, local 20.225-22.992 inside Njh8_zq9_DM-8835, with the new actual-source crop wieniawski-scherzo-tarantelle-staff4-eb-eb-c-eb-eb-eb-verified.png. The truth manifest now has two live accepted Staff 4 source/audio phrases and four verified positive source phrases. The right-2 seven-note lane remains the next audit target; it cannot inherit acceptance from the six-note prefix.

The remaining M1 work is continuing the low-cost media pass over every unmeasured strict-ledger video in chronological windows, then replacing the sample-biased estimate with full-archive checked intervals. The remaining M10 work is the larger benchmark suite for repeated notes, arpeggios, fast runs, score-free exercises, score-coordinate truth, accepted notation/media completeness, and practice-time accounting regressions.

The long-phrase track should now run in parallel with full-archive scanning. Waiting for all 234 h 57 min of archive coverage before extending one musical lane would be too slow. The faster strategy is a narrow accepted-measure lane:

1. keep the accepted 2026-05-03 Staff 4 source/audio phrase as the anchor;
2. keep the corresponding Scherzo-Tarantelle MusicXML measure-window and actual-score crop fixed as source truth;

3. search adjacent stored and source-rescanned detected note series against that same source lane;
4. accept the expansion only if the exact score notes, displayed transcription, local audio, local video, source crop, and truth gate all agree;
5. extend outward one verified adjacent note group at a time until the lane becomes a phrase.

The next coding pass should:

1. promote the accepted six-note Staff 4 lane into the visible compact grouped score–clip–audio–transcription pair after live deploy verification;
2. generate or update the next Staff 4 right-2 audit packet for Eb5 Eb5 C5 Eb5 Eb5 Eb5 C5;
3. compare the right-2 source sequence, detected sequence, source crop range, pitch trace, and local audio/video before any acceptance;
4. accept a longer compact grouped score–clip–audio–transcription pair only if the expanded exact MIDI sequence, paired audio, source crop, and truth gate all pass;
5. keep reporting the first blocking note when an expansion fails so the next extraction attempt has a precise target;
6. keep running queue-only owner-media sync in parallel, starting with violin 1 at *2250–2340;
7. rerun the active-practice scan after those uploads and record checked, active, and pending totals.

Curtis now has the first verified measure-window before M1 is complete, but it still cannot claim a longer score match from guessed scanned crops, one-note anchors, reference-audio pitch traces, or a mismatched adjacent audio note. The speed-up is to keep the accepted Staff 4 lane fixed and extend it only when each added source note passes the same gates that a future long phrase will need.

6 Materials and methods

Backend

1784

The backend is a FastAPI service deployed on Railway. It exposes <https://curtis.aolabs.io/api/curtis/media-status> for compact state, <https://curtis.aolabs.io/api/curtis/ops-check> for fuller operational state, local stored sample media for embedded playback after the media is violin-positive, sample indexes for duplicate-window avoidance, and run endpoints for scans, media probing, analysis, and coaching. Persistent runtime state stores sources, inventory, review findings, media samples, analysis runs, and scan history.

Source scan

1791

The default public source is <https://www.youtube.com/@nalalan>. The scanner queries public video metadata, classifies videos by title, duration, and candidate reasons, and stores inventory records. Dated practice logs and long-form videos are treated as practice candidates. The media probe now keeps one early baseline window but then probes deeper quarter, midpoint, five-eighths, three-quarter, and tail windows before spending additional passes on early setup probes, and each stored sample is keyed by video identifier plus window start so later passes can expand coverage without overwriting earlier evidence. When configured above the dense-sampling threshold, the probe and owner-media helper generate dense 90-second windows from the start of each video through the latest valid window so the full practice archive can be checked without the earlier retention cap. Each sample is then classified by the violin-presence sampler before it can count as active practice evidence. The owner-media helper also checks its local cache of already captured browser samples, reuses classifier results when the file signature and classifier version match, prioritizes active-scan pending windows before generic expansion or blind sampling, can run in queue-only mode to consume chronological pending windows without falling back to unrelated cached positives or blind windows, resumes cached queue windows after an interrupted capture, includes cached negative queue windows so non-playing time is checked rather than ignored, prioritizes unsynced violin-positive cached windows during normal mode before capturing more blind windows, and expands around already-found violin-positive windows before returning to blind sampling. The practice-hours total uses a stricter title-confirmed ledger from **violin** 1 onward: violin-numbered videos and date-titled logs count toward the top-line session duration, while unrelated long videos remain indexed but outside the practice-hours total. The scan does not itself inspect performance quality.

Model review

The model layer reviews sampled media sections where usable excerpts exist. GPT-5.5 is used for text and visual review, and separate audio models are configured for audio review and piece-title verification. Findings are sanitized so weak evidence terms such as obscured, not audible, or no clear evidence produce an unjudged state rather than an unsupported critique. Piece labels are also sanitized: coach reviews can store candidate evidence but cannot create confirmed repertoire titles, and the piece-identification layer must provide source-window evidence plus independent verification before it updates the current piece list. Same-video consensus can compare multiple sampled windows before accepting a title; for long date-stamped practice captures with later samples, setup or non-playing windows are excluded from confirmation, and confirmed active titles are versioned so older labels are rechecked when source rules change. User-rejected false labels are excluded from later passes only for the source where the correction applies, so later videos can still identify that repertoire when present. A source correction also invalidates older confirmed labels for that same source instead of promoting the next stale candidate, and repeated corrected false labels keep that source out of the confirmed repertoire list until a stronger acceptance path exists. Rejected proposed titles, top candidates, and verification titles are scrubbed from current API output, not only from confirmed piece records. The 5/1 forensic pass rejected prior Paganini, Bruch, Mendelssohn, Brahms, Sibelius, Tchaikovsky, Wieniawski, Saint-Saens, Ravel, Bazzini, Ernst, Carmen, Zigeunerweisen, Bach Partitas, Kreisler Praeludium and Allegro, Sarasate family labels, Ysaye Sonata No. 3 Ballade, Mozart K. 216, Glinka Ruslan and Lyudmila, Strauss Till Eulenspiegel, Strauss Don Juan, Smetana Bartered Bride, Prokofiev Classical Symphony, Dvorak Carnival Overture, Shostakovich Symphony No. 5, Ravel Bolero, the first broad orchestral candidate batch, and the later overture, dance, and pops-orchestral candidate batch for that source. Alan then supplied the ground-truth source label: Haydn Symphony No. 94, IV. Finale, Violin I part. The system now treats that label as human-confirmed for 5/1, stops injecting new seeded guesses for the same source, and keeps Curtis-level completion unscored until judged playing evidence supports a score. Alan also supplied the 5/2 source label, Wieniawski Scherzo-Tarantelle, Op. 16, and then confirmed that all violin footage on 5/3 was the same piece. The training state now records all three source labels as confirmed sources and reports zero score-aligned windows until pitch/rhythm extraction has matched a source window to a score section. Transcribed samples can now produce learned pitch/rhythm fingerprints only after their media window is violin-positive, and visible notation requires note/rhythm verification. Future samples can receive a nearest-source hint

when their extracted notes and rhythms resemble a prior confirmed source, an explicit title-labeled calibration upload, a symbolic reference, or a public labeled YouTube reference indexed through the official Data API; if the extracted pattern does not belong to a known score, it remains piece-or-exercise pending rather than being forced into repertoire. Public reference metadata can improve candidate vocabulary and target selection, but it does not become an audio fingerprint until a permitted media source exists. Possible labels and model-only verified titles remain uncertain daily evidence unless a source label is confirmed, so future misses appear as missing evidence rather than as wrong repertoire. Piece-identification output does not create a Curtis-level readiness percentage by itself; readiness scoring requires judged playing evidence. Dated practice entries are keyed by the source-video date rather than the later processing time, so historical uploads cannot masquerade as today's practice state. The owner-browser sync uses the sample index to prioritize uncaptured public YouTube windows rather than re-uploading already reviewed excerpts, and the piece-identification pass can test multiple active moments inside each captured sample before withholding the title. The daily-record builder joins the strict violin 1 ledger to violin-positive media samples, hidden machine note/rhythm events, practice-time evidence, clips, score packet support, repeated-fragment or repeated-pattern heat maps, score-aware read fields, and observation records. Repertoire entries are built only from confirmed daily-record evidence and keep progress percentages unassigned until clips, notation, and score or pattern alignment support them.

Progress ledger

AO Progress tracks Curtis as part of a larger multi-application state record. The first progress snapshot included Curtis home, Curtis media state, other AO Labs apps, Imagineer state, Imagineer paper, Relay state, and public project surfaces. Curtis contributes dated practice inventory, current focus, review status, and future trend signals.

7 Limitations

The current system cannot predict acceptance at Curtis. It also cannot replace a musician, teacher, jury, studio lesson, or audition process. It can be wrong when the source media are incomplete, when audio is unavailable, when video frames hide the physical action, or when model review fails. The broad micro-transcription gate is currently held closed: candidate note runs can be stored, but no displayed passage transcription should be treated as solved until its notes audibly match the paired clip. The only current displayed notation is two strict audio-checked note fragments, so the useful claim is limited: Curtis Media Review turns practice media into a persistent, source-bounded record that can support iteration over time.

References

1. K. A. Ericsson, R. T. Krampe, C. Tesch-Romer, The role of deliberate practice in the acquisition of expert performance. *Psychological Review* **100**, 363–406 (1993).
2. A. Williamon, Ed., *Musical Excellence: Strategies and Techniques to Enhance Performance* (Oxford University Press, Oxford, 2004).
3. G. Klickstein, *The Musician's Way: A Guide to Practice, Performance, and Wellness* (Oxford University Press, New York, 2009).
4. A. N. Pham, *Curtis Media Review*, Live software system, curtis.aolabs.io, 2026.
5. J. P. Bello *et al.*, A tutorial on onset detection in music signals. *IEEE Transactions on Speech and Audio Processing* **13**, 1035–1047 (2005).
6. A. de Cheveigne, H. Kawahara, YIN, a fundamental frequency estimator for speech and music. *The Journal of the Acoustical Society of America* **111**, 1917–1930 (2002).

7. E. Benetos, S. Dixon, D. Giannoulis, H. Kirchhoff, A. Klapuri, Automatic music transcription: challenges and future directions. *Journal of Intelligent Information Systems* **41**, 407–434 (2013). 1888
8. M. Muller, *Fundamentals of Music Processing: Audio, Analysis, Algorithms, Applications* (Springer International Publishing, Cham, 2015). 1890
9. E. J. Humphrey, J. P. Bello, Y. LeCun, presented at the Proceedings of the 13th International Society for Music Information Retrieval Conference, pp. 403–408. 1892
10. A. N. Pham, *AO Progress Ledger*, Live longitudinal state system, progress.aolabs.io, 2026. 1894
11. Google for Developers, *YouTube Data API* (2026; <https://developers.google.com/youtube/v3>). 1895
12. W3C Music Notation Community Group, *MusicXML* (2026; <https://www.musicxml.com/>). 1897
13. International Music Score Library Project, *IMSLP: Petrucci Music Library* (2026; <https://imslp.org/>). 1898